Minimum State Products on Dijkstra's Token Ring: Exact Values for $3 \le n \le 9$ and a Large-n Conjecture Supported through $n = 10$

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v1.0, April 24, 2026

Abstract

Headline. We determine the exact minimum state product M_n for $n \in \{3, \dots, 9\}$ in Dijkstra's central-daemon model, unconditional at $n \in \{3, 4\}$, and conditional for $n \in \{5, \dots, 9\}$ on the replayable lower-bound certificates of appendix C, and conjecture $M_n = 4 \cdot 3^{n-2}$ for all $n \ge 9$. Between $n = 8$ and $n = 9$ the ratio M_n/M_{n-1} breaks from 3 to 27/8: a small-n absorber family saturates the bound through $n = 8$, the ternary-strip family saturates at $n = 9$, and no witness from either family extends past its crossover point (Observation 6 and theorem 7).

Evidence. The upper bound is realized through $n = 10$ by a Lean 4 / Mathlib formalization of the ternary-strip witness CUP-2, with a Python-verified sibling construction (CLB) extending through $n = 18$. The lower bound is backed through $n = 9$ by the replayable exhaustive-search procedure of appendix C, with worked-example certificates shipped at small n and the full per-orientation stream regenerable via the shipped driver. Structural detectors on a 97-record stratified corpus separate valid witnesses from sub-threshold candidates: a circulation linear program on the lifted-defect graph separates 93 of 97 records, and a strict upgrade, the *sink kernel* of the forced-move digraph on non-good configurations (Axis C), a structural certificate that rules out valid rule-table completions of candidate cycles, separates all 97. Sink-kernel nonemptiness on every tested sub-threshold record combined with a general monotonicity lemma in detOf yields a one-sided sub-threshold certificate on those records. Under the relaxed-connectedness convention of the 1985 Stanford seminar [8], an SMT encoding newly certifies $M_5^{\mathrm{rel}} = M_5 = 96$ (§ 3.4), the first case at $n \ge 5$ in which $M_n^{\mathrm{rel}} = M_n$.

What remains. A uniform-in-n analytical lower bound is open; the mechanism of the ratio crossover at $n = 9$ is open (the 1985 ARG LCM bound is ruled out on both scope and insensitivity grounds; see § 5.3; the handle "ARG" comes from the unsigned Stanford seminar notes [8, §2–§3]); transport of the large-n bounds to the relaxed-connectedness convention is open beyond $n = 5$. A systematic catalog of failed analytical routes (§ 7) locates the obstruction at the joint interaction of the good cycle, its mover sequence, and the determined rule-table entries.

1 Introduction

1.1 Dijkstra's self-stabilization and the product-minimization question

In 1974 Dijkstra [4] posed a now-canonical problem in distributed computing: design n ring processors, each with only local visibility of its two neighbors, whose joint dynamics under a non-deterministic central daemon converges to mutual exclusion from *any* initial configuration. A processor P_i has m_i states and a local transition function $f_i : \mathrm{Fin}(m_{i-1}) \times \mathrm{Fin}(m_i) \times \mathrm{Fin}(m_{i+1}) \to \mathrm{Fin}(m_i)$. P_i is *privileged* at a configuration $c = (c_0, \dots, c_{n-1})$ when $f_i(c_{i-1}, c_i, c_{i+1}) \ne c_i$. A valid *self-stabilizing* system satisfies six properties: liveness, mutual exclusion, closure, convergence, fairness, and legitimate-state connectedness (defined precisely in § 2).

Dijkstra exhibited three solutions: a K-state uniform construction ($K \geq n$, product K^n; Dijkstra's original statement uses $N+1$ machines and requires $K > N$, which becomes $K \geq n$ for an n-processor ring), a 4-state non-uniform construction (with the post-1985 caveat that as printed it solves the line, not the ring; see [8]), and a 3-state uniform construction with product 3^n. In the winter 1985 Stanford problem seminar [8], Knuth posed the optimization question exhibited here as our main problem: minimize the state-count product $m_0 m_1 \cdots m_{n-1}$ over all valid systems on n processors. Write M_n for that minimum. The 1985 seminar pinned M_n into the gap $2^n < M_n \leq 3^n$ for $n > 4$, with $M_n^{1/n}$ growth rate unsettled. Knuth recorded two specific asymptotic questions: is $\limsup_n M_n^{1/n} < 3$, and is $\liminf_n M_n^{1/n} > 2$? Forty years later both are still open.

1.2 Contributions of this paper

Results.

1. Exact values of M_n for $n \in \{3, \ldots, 9\}$ in the central-daemon model ($n \in \{3, 4\}$ unconditional; $n \in \{5, \ldots, 9\}$ conditional on the replayable lower-bound certificates of appendix C), with the upper bound Lean-checked through $n = 10$ and the lower bound established by exhaustive computation through $n = 9$.

2. Non-extension of each witness family past its crossover point: the small-n absorber family $(2^3, 4, 3^{n-4})$ fails at every $n = 9$ multiset, and the ternary-strip family sits at M_n only at $n = 9$ (theorem 7). The arithmetic M_n/M_{n-1} break from 3 to 27/8 is then a corollary of the exact values.

3. Ruling out ARG's 1985 LCM bound[1] as the mechanism of the $n = 9$ crossover: ARG's hypothesis excludes every adjacent-binary witness that realizes M_n for $n \in \{4, \ldots, 8\}$, and ARG's LCM functional is constant or strictly nested between $n = 8$ and $n = 9$ on the non-adjacent orientations it does cover (§ 5.3).

4. The conjecture $M_n = 4 \cdot 3^{n-2}$ for $n \geq 9$. It is established at $n = 9$ (theorem 5); at $n \in \{10, 11\}$ the upper bound is Lean-checked (through $n = 10$) or Python-verified CLB (through $n = 18$) and the lower bound is supported by Axis C per-probe but *not* proved per-multiset: all 320 tested $n = 10$ sub-threshold probes fire Axis C on every (291-multiset) sub-threshold class, and all 585 tested $n = 11$ probes fire across 563 of 564 sub-threshold multisets (one SMT-budget UNKNOWN); per-probe firing under the DFS + SMT pipeline does not by itself rule out every candidate good cycle on every multiset, see the scope caveat of § 6.5. At $n \geq 12$ the conjecture is a conjecture: the formula $4 \cdot 3^{n-2}$ is realized as an *upper* bound through $n = 18$ (Lean through $n = 10$, Python-verified CLB through $n = 18$), making it a plausible candidate target for the lower bound, but the only direct lower-bound evidence past $n = 11$ is extrapolation from corpus-level detector separation on $n \in \{5, \ldots, 10\}$ and the two full-layer sweeps at $n = 10, 11$. If the conjecture is true it forces $M_n^{1/n} \to 3$ and would resolve a connected-model variant of Knuth's 1985 asymptotic questions (corollary 11); the unqualified 1985 questions were posed under the relaxed-connectedness convention and their resolution requires additional transport results (§ 8.8).

5. A z3 SMT certificate that $M_5^{\mathrm{rel}} = M_5 = 96$ (theorem 12): none of the five $n = 5$ sub-threshold multisets admits a relaxed-valid rule table. This is the first case at $n \geq 5$ in which $M_n^{\mathrm{rel}} = M_n$; at $n = 6$, seven of twelve sub-threshold multisets are UNSAT-certified with the remaining five returning z3 UNKNOWN within a 3600 s budget per multiset (§ 8.8).

[1] "ARG" is the unsigned handle under which the bound appears in the 1985 Stanford seminar notes [8, §2–§3]; we use the seminar's label throughout.

6. A corpus-level "currency null" result (Observation 20): on the 97-record stratified corpus, no dynamical currency (mover-pattern period T_μ, detOf-coverage κ, entropy H) separates valid from sub-threshold more cleanly than $\prod m_i$ does, and at $n = 9$ every candidate dynamical currency *flips direction.* This rules out the natural reformulation "lower-bound a different quantity."

Methods.

7. The CLB ternary-strip witness family on the state vector $(2, 3, \ldots, 3, 2)$, Python-verified for $n \in \{5, \ldots, 18\}$.

8. A second ternary-strip witness CUP-2 on the same state vector, Lean 4 / Mathlib formalized for $n \in \{4, \ldots, 10\}$.

9. Structural detectors on candidate good cycles: the circulation-LP detector separates 93 of 97 records in a stratified corpus (the four failures at the $n = 9$ non-adjacent $\{2^3, 3^5, 4\}$ orientations); the sink-kernel upgrade (Axis C, § 6.5) separates all 97 on the tested corpus and, combined with a general detOf-monotonicity lemma, supplies a one-sided sub-threshold certificate on each tested record. The corpus spans $n \in \{5, \ldots, 10\}$ on both sides; the $n = 10$ sub-threshold side is further confirmed by an extended two-tool (DFS + SMT) sweep that fires Axis C on every probe the pipeline surfaced, 320/320 probes across every one of the 291 $n = 10$ sub-threshold multisets (this is a per-probe statement, not a per-multiset completeness claim; see the scope caveat in § 6.5).

A systematic diagnostic catalog of failed analytical lower-bound approaches across five invariant families (§ 7) accompanies these contributions as a service to the community.

1.3 Why this paper

In 1985, Knuth [8] asked how small the product of state-counts can be on a self-stabilizing ring. Forty years later, the question is still open in both directions: Dijkstra's 3^n is the best known upper bound, and the 1985 seminar's 2^n is the best known lower bound. This paper closes the gap exactly for $n \leq 9$ and proposes $4 \cdot 3^{n-2}$ as the answer for $n \geq 9$.

Why now. Four tooling shifts make the $n \leq 9$ frontier tractable in 2026 in a way it was not in 1985 or 2005. First, modern SMT solvers dispatch rule-table feasibility on sub-threshold multisets in seconds to hours where exhaustive rule-table enumeration requires 10^{10}–10^{12} candidates per multiset (§§ 3.4 and 8.8). Second, Lean 4 / Mathlib now supports finite per-n rank-table convergence certificates sealed by `native_decide` at state-counts up to 26244 (appendix A). Third, consumer compute makes exhaustive lower-bound search at $n = 9$ (tens of thousands of multiset/orientation classes) a hours-to-days job rather than a decades job. Fourth, and what changes the feasible *shape* of the inquiry rather than only its speed, frontier LLMs (Claude Opus, GPT-5.4) run as specialist research agents under a residue-prompt protocol (`docs/residue_prompt_v2.md`) and an RA/PA/LE division of labour (`docs/ra_le_workflow.md`), letting a single human orchestrator run, log, and cross-check the probe volume required to chart the failed-approach landscape catalogued in § 7 and appendix D: $\sim$2,100 Python research probes ($\sim$633,000 lines) carrying 721 logged explorations across 30 chronological logs, plus the $\sim$29,000-line Lean formalization and the $\sim$79,000 lines of Lean `attic/` preserving failed routes. The probe volume is load-bearing: the detector hierarchy of § 6.5 and the obstruction catalog of § 7 both require exhausting a *space* of candidates, not verifying a single one, and that space could not have been swept at this density by unassisted human enumeration in the available calendar time. None of these four shifts was available to the 1985 seminar; none has previously been applied to this problem.

To our knowledge, no prior work has determined M_n exactly beyond M_4 in the central-daemon model; identified the sharp ratio crossover at $n = 9$ (in the sense of remark 8); produced machine-verified constructions at this scale; documented a systematic analytical-obstruction catalog; or produced structural detectors separating the $n \leq 10$ sub-threshold corpus from the valid-witness corpus with a one-sided certificate via sink-kernel monotonicity on the forced-NG digraph (§ 6.5).

1.4 Companion repository

This paper ships alongside a companion repository (release tag and Zenodo DOI to be pinned at submission; see appendix C.11). All Lean modules, Python probes, the exhaustive-search driver, SMT encodings, and data artifacts referenced in the body are in that tree; the paths indexed in appendix F are relative to the release tag. The repo root contains `REVIEWER.md`, which maps each theorem and conjecture in this paper to a single reproduction command. Toolchain versions (Lean 4, Mathlib revision, Python, z3) are pinned: Lean and Mathlib via `lean-toolchain` and `lakefile.lean`; Python and z3 via `requirements.txt`; the pinned toolchain rebuilds every claim end-to-end from a clean clone. The reader should treat the paper and the repo as halves of a single object.

1.5 Paper structure

§ 2 fixes notation and the model. § 3 states theorems 5 and 7 and Observation 6 and Conjectures 9–10. § 4 presents the upper-bound constructions, § 5 presents the exact lower bounds and the phase-transition certification, and § 6 describes the circulation detector and the empirical evidence behind the large-n conjecture. § 7 catalogs the failed analytical approaches, § 8 lists open questions, § 9 surveys related work, and § 10 concludes. Appendices contain Lean theorem statements, the formal construction of the detector, the exhaustive-verification methodology with replayable certificates, extended discussion of the failed approaches, and small-n witness data.

2 Preliminaries

2.1 The ring model

A *ring* consists of n processors $P_0, \ldots, P_{n-1}$ with state counts $\mathbf{m} = (m_0, \ldots, m_{n-1})$, each $m_i \geq 2$, and local transition functions

$$f_i : \mathrm{Fin}(m_{i-1}) \times \mathrm{Fin}(m_i) \times \mathrm{Fin}(m_{i+1}) \to \mathrm{Fin}(m_i), \qquad i \in \{0, \ldots, n-1\},$$

with indices modulo n. A *configuration* is a tuple $c = (c_0, \ldots, c_{n-1}) \in \mathrm{Config}(\mathbf{m}) := \prod_i \mathrm{Fin}(m_i)$. Processor P_i is *privileged* at c when $f_i(c_{i-1}, c_i, c_{i+1}) \neq c_i$. A *move* from c at a privileged P_i replaces c_i by $f_i(c_{i-1}, c_i, c_{i+1})$ and leaves the other coordinates unchanged. Under the *central daemon* the next-step move is one privileged P_i chosen non-deterministically.

2.2 Model convention

Convention. Throughout this paper, *validity* means Dijkstra's central-daemon model with the six properties of § 2.3 (liveness, mutual exclusion, closure, convergence, fairness, legitimate-state connectedness). The connectedness property is the one carried over from Dijkstra's original 1974 paper [4, §3], which required that for any two legitimate states there exist a sequence of moves transferring one to the other. Under this requirement the legitimate states form a single orbit under the unique good move, so the Gray-code multi-cycle constructions studied in the 1985 seminar do not qualify.

This is a model choice, not a correction. Historical references to $M_4 = 16$ [8, p. 70] should be read under the *relaxed-connectedness convention* of the 1985 seminar (credit due to Knuth), in which Dijkstra's legitimate-state connectedness requirement was dropped and the good configurations were allowed to split into several disjoint good cycles; in the central-daemon model of this paper (which retains Dijkstra's connectedness), $M_4 = 24$ (theorem 5). Later references to the same convention use *relaxed-connectedness* or *1985 convention* without further qualifier.

2.3 The six validity properties

A system $(\mathbf{m}, f_0, \ldots, f_{n-1})$ is *valid* when:

(1) **Liveness.** In every configuration at least one processor is privileged.

(2) **Mutual exclusion.** In every *good* configuration (see below) exactly one processor is privileged.

(3) **Closure.** The unique move from a good configuration leads to another good configuration.

(4) **Convergence.** No daemon schedule starting from any bad configuration cycles back to itself; equivalently, every execution reaches a good configuration in finitely many steps.

(5) **Fairness.** Every cycle of moves through good configurations fires every processor at least once.

(6) **Legitimate-state connectedness.** For any two good configurations g, h there is a sequence of good moves leading from g to h. Equivalently (given the preceding properties), the good configurations form a single directed cycle under the unique good move.

Configurations are partitioned by the designer into *good* (mutual exclusion holds, the designer's certificate of correct service) and *bad* (transient). Property (6) is Dijkstra's original legitimate-state connectedness requirement [4, §3]. The 1985 Stanford seminar [8, p. 70] worked with a relaxed version of property (6) in which the legitimate states were allowed to split into several disjoint good cycles; under that convention the $M_4 = 16$ Gray-code construction (which splits the 16 legitimate states into two disjoint 8-cycles) is admissible. This paper retains Dijkstra's original connectedness requirement, under which $M_4 = 24$ (theorem 15).

2.4 Dijkstra's three solutions

Dijkstra's K-state uniform construction [4] achieves product K^n for $K \geq n$ (Dijkstra's original statement uses $N+1$ machines and requires $K > N$; translating to a ring of $n = N+1$ processors gives $K \geq n$). His 4-state non-uniform construction achieves product 4^{n-1} on a line; as the 1985 seminar documented [8, p. 76], this construction does not close into a ring without modification. Dijkstra's 3-state uniform construction (Solution 3) achieves product 3^n, with the convergence proof appearing only in 1986 [5]. Solution 3 is the only one of the three whose product matches the trivial uniform $M_n^{1/n} \to 3$ asymptotic without further analysis.

2.5 The state-product minimum M_n

We write

$$M_n := \min\Big\{\textstyle\prod_{i=0}^{n-1} m_i \,:\, (\mathbf{m}, f_0, \ldots, f_{n-1}) \text{ is valid in the sense of § 2.3}\Big\}$$

for $n \geq 3$, where validity includes the legitimate-state connectedness property (6). The connected-model analogues of Knuth's two open questions [8, p. 68] ask whether $\limsup_n M_n^{1/n} < 3$ and whether $\liminf_n M_n^{1/n} > 2$; the original 1985 questions were posed under the relaxed-connectedness convention of § 2.2, and transporting any bound from the connected model to that convention is an open problem (§ 8.8).

2.6 Two distinct length quantities

Two quantities associated with the good cycle of a valid system both deserve to be called "length." We will keep them rigidly distinct:

Definition 1. The *good-configuration count* L is the number of distinct configurations on the good cycle. The *mover-pattern period* T_μ is the period of the cyclic sequence of firing processor indices around the good cycle.

These are not in general equal. Both ternary-strip constructions of § 4 have $T_\mu = 3n - 2$, but the CLB construction has $L = n^2 - 2n + 8$ while the CUP-2 construction has $L = (n+2)(n+3)/2 - 5$.

2.7 Sub-threshold terminology

We will repeatedly compare a candidate state-count vector against the proved or conjectured target product at each n.

Definition 2. For $n \geq 5$ define

$$B_n := \begin{cases} 32 \cdot 3^{n-4} & \text{if } 5 \leq n \leq 8 \quad \text{(the proved exact value } M_n\text{)}, \\ 4 \cdot 3^{n-2} & \text{if } n \geq 9 \quad \text{(the proved exact value } M_n \text{ at } n = 9;\ \text{the large-}n \text{ conjectured target).} \end{cases}$$

Definition 3. A *detector record* is a pair $(\mathbf{m}, gc)$ where $\mathbf{m}$ is a state-count vector and gc is a candidate good cycle on $\mathbf{m}$ (a closed sequence of distinct configurations with exactly one privileged processor at each step and fairness). For *valid-system records*, gc is the good cycle of a verified valid system. For *candidate records* no valid extension is assumed.

Definition 4. A detector record $(\mathbf{m}, gc)$ is

- *sub-threshold* if $\prod_i m_i < B_n$;
- a B_n*-threshold witness* if it is a valid-system record with $\prod_i m_i = B_n$ (proved sharp only when $B_n = M_n$ has been established);
- a *valid super-threshold witness* if it is a valid-system record with $\prod_i m_i > B_n$.

We avoid the unqualified phrase "sharp valid witness" for any record where the lower bound has not been proved.

2.8 Structural objects used later

The lower-bound discussion of §§ 6 and 7 requires several derived objects, defined here in summary form (full Lean-level definitions are in the source repository under `LeanMn/LowerBound/`, with a glossary in the supplementary lower-bound history memo listed in appendix F).

- Given a candidate good cycle $gc = [c_0, \ldots, c_{L-1}]$, the *determined dictionary* $\mathrm{detOf}(gc)$ is the partial rule table read off the cycle: for each context (i, l, s, r) realized at some step k, $\mathrm{detOf}(gc)$ records $c_{k+1}(i)$. Stay-entries (where $c_{k+1}(i) = c_k(i)$) and move-entries (where they differ) are tracked separately.
- $\mathrm{NG}(gc)$ denotes the set of *non-good configurations* (those not appearing on gc).
- The *Hamming-1 tube* $T_{N_1}(gc)$ is the set of configurations at Hamming distance exactly 1 from some c_k that are also *value-consistent* with gc (every coordinate value appears at the corresponding position somewhere on gc).

- The *forced graph* on a set $S \subseteq \mathrm{NG}(gc)$ has edges $c \to c'$ whenever c has a forced successor $c' \in S$ under $\mathrm{detOf}(gc)$.

- The *sink kernel* $\mathrm{SK}(gc)$ is the fixed point of iteratively removing sinks (vertices with no out-edge) from the forced graph on $\mathrm{NG}(gc)$.

- The *lifted-defect graph* on gc has vertex set $V_{\mathrm{lift}}(gc) = \{(k, q, a) : a \neq c_k(q),\ c_k[q := a] \in T_{N_1}(gc)\}$ with the edge taxonomy of § 6.

3 Main results

Table 1 summarizes the four independent scales at which the results of this section are grounded; every later subsection is a claim made at one of these scales.

Table 1: Scales of evidence. "Established" means an exhaustive, replayable certificate exists; "Lean-checked" means a sorry-free Lean 4 / Mathlib proof on finite-per-n tables discharged by `native_decide`; "Python-verified" means `verify_system` passes on the stored witness; "Axis C-supported" means the sink-kernel detector fires on every tested sub-threshold record. The four columns anchor every later reference.

scale	what it establishes	range
exhaustive LB certificate	$M_n \geq B_n$	$n \in \{3, \dots, 9\}$
Lean-checked UB witness	$M_n \leq 4 \cdot 3^{n-2}$ (ternary-strip)	$n \in \{4, \dots, 10\}$
	$M_n \leq 32 \cdot 3^{n-4}$ (absorber)	$n \in \{5, 6, 7, 8\}$
Python-verified UB witness	$M_n \leq 4 \cdot 3^{n-2}$ (CLB)	$n \in \{5, \dots, 18\}$
Axis-C sub-threshold certificate	$M_n \geq B_n$ on tested records	$n \in \{5, \dots, 10\}$

3.1 Exact values

Theorem 5 (Exact values, central-daemon model). *In Dijkstra's central-daemon model of § 2.3, the minimum state product takes the values*

n	M_n	*factorization*
3	8	2^3
4	24	$2^3 \cdot 3$
5	96	$2^5 \cdot 3$
6	288	$2^5 \cdot 3^2$
7	864	$2^5 \cdot 3^3$
8	2592	$2^5 \cdot 3^4$
9	8748	$2^2 \cdot 3^7$

Each row combines an upper-bound witness with a lower-bound certificate (§§ 4 and 5). The upper bounds for $n \in \{4, \dots, 10\}$ are Lean-checked (theorems 13 to 15). The lower bounds are certificate-backed as follows: $n = 3$ by trivial enumeration (no sub-threshold multisets exist); $n = 4$ analytically by theorem 15; $n \in \{5, \dots, 8\}$ by the exhaustive-search procedure of appendix C, whose driver is shipped and whose per-orientation rejection stream is pre-generated for $n \in \{3, 4\}$ and the $n = 5$ all-binary orientation, and regenerable from a clean clone for any other orientation (appendices C.10 and C.11); and $n = 9$ by the targeted absorber-specific computational probes of § 5.2 on the 3 multisets at product 7776 and the 24 multisets at products in $(7776, 8748)$. The rows $n \in \{4, \dots, 9\}$ are sharp; sharpness at $n = 10$ is conjectural (§ 3.3).

Certificate-backing status. The $n = 3$ row is unconditional (trivial enumeration); $n = 4$ is unconditional via theorem 15. For $n \in \{5, \ldots, 9\}$, the $\geq$ direction of theorem 5 is backed by the exhaustive search of appendix C. The deterministic coverage manifest (C1 + C2, overall hash `f684ed53216a0d7f`) ships for every $n \in \{3, \ldots, 9\}$ under `artifacts/rejections/` in the release tag (appendix C.11); the per-certificate rejection stream ships in full for $n \in \{3, 4\}$ and for the all-binary orientation at $n = 5$ as a worked example of the search mechanism. Per-cert streams for the remaining $n = 5$ orientations and for all $n \in \{6, \ldots, 9\}$ are regenerable from a clean clone with the driver of appendix C.10 (the full stream at worst-case multisets is prohibitive in disk size, which is why it is not pre-generated); see the reproduction map in `REVIEWER.md` (§ 1.4).

3.2 The ratio crossover at $n = 9$

Given theorem 5, the ratio identity is arithmetic.

Observation 6 (Ratio crossover)**.** Conditional on the same replayable certificates as theorem 5, the consecutive ratios are

$$\frac{M_5}{M_4} = 4, \qquad \frac{M_n}{M_{n-1}} = 3 \text{ for } n \in \{6, 7, 8\}, \qquad \frac{M_9}{M_8} = \frac{27}{8}.$$

Both endpoints of the constant-3 plateau are structural: $M_5/M_4 = 4$ is the absorber family turning on (quaternary enters at $n = 5$), and $M_9/M_8 = 27/8$ is the absorber family turning off.

The non-trivial statement is that neither witness family realizes M_n past its own crossover point: the small-n absorber multiset $(2^3, 4, 3^{n-4})$ (§ 4.2) fails at every $n = 9$ multiset, and the ternary-strip formula $4 \cdot 3^{n-2}$ is sharp only at $n = 9$ among currently proved exact values.

Theorem 7 (Non-extension of each family)**.** *Conditional on the replayable lower-bound certificates of appendix C:*

1. *For every multiset* $\mathbf{m}$ *at* $n = 9$ *with* $\prod m_i = 32 \cdot 3^{9-4} = 7776$*, namely the three small-*n *absorber shapes* $\{2^3, 3^5, 4\}$, $\{2^4, 3^4, 6\}$*, and* $\{2^5, 3^3, 9\}$ *(table 10), no valid rule table exists (§ 5.2).*
2. *The ternary-strip product* $4 \cdot 3^{n-2}$ *is sharp at* $n = 9$ *(theorem 5) and realized by verified witnesses at* $n \in \{4, \ldots, 10\}$ *via CUP-2 (theorem 14); at* $n \leq 8$ *it is strictly above* M_n *by a factor of* $\frac{4 \cdot 3^{n-2}}{32 \cdot 3^{n-4}} = 9/8$.

Sharpness of the ternary-strip product at $n = 10$ *is conjectural.*

Remark 8 (Terminology)**.** We informally refer to the crossover of Observation 6 and theorem 7 as a *phase transition*. The term is used only in this informal sense; we do *not* claim to have identified a structural mechanism that *causes* the break at $n = 9$ specifically. A uniform mechanistic explanation is open (§ 5.3), and in particular ARG's 1985 LCM bound does not account for it.

3.3 The large-n conjecture

Conjecture 9 (Upper bound, all $n \geq 9$)**.** *For every* $n \geq 9$ *there exists a valid rule table on the ternary-strip state vector* $(2, 3, 3, \ldots, 3, 2)$*, giving the upper bound* $M_n \leq 4 \cdot 3^{n-2}$*. The CLB and CUP-2 rule tables of § 4 are two candidate families realizing this state vector in the verified finite ranges.*

Status: Lean-checked for $n \in \{4, \ldots, 10\}$ via CUP-2 (theorem 14); Python-verified for $n \in \{5, \ldots, 18\}$ via the CLB construction. For $n \geq 19$ only Dijkstra's classical $M_n \leq 3^n$ [4] is currently available, a factor $9/4 \approx 2.25$ looser than the conjectured $4 \cdot 3^{n-2}$.

Conjecture 10 (Lower bound, all $n \geq 9$)**.** *Every valid system on $n \geq 9$ processors has $\prod_i m_i \geq 4 \cdot 3^{n-2}$.*

Status of Conjecture 10, stratified by regime:

- $n = 9$ is **established** by exhaustive computation (theorem 5 and § 5.2): every sub-target product class is eliminated by a replayable certificate shipped with the release tag (appendix C.11).
- $n \in \{10, 11\}$ is **Axis C-supported, not proved**: no exhaustive certificate search has been run, but the DFS + SMT sweep fires Axis C on all 320 tested probes across the 291 sub-threshold multisets at $n = 10$ and on all 585 probes across 563 of 564 sub-threshold multisets at $n = 11$ (one SMT-budget UNKNOWN), which eliminates each of them under sink-kernel monotonicity *on the tested records* (§ 6.5).
- $n \geq 12$ is **conjecture**. Supporting evidence is (a) two Lean- and Python-verified witness families realizing $4 \cdot 3^{n-2}$ (through $n = 10$ and $n = 18$ respectively), and (b) corpus-level detector separation on 10 valid witnesses and 87 sub-threshold candidates spanning $n \in \{5, \ldots, 10\}$, extended by two full-layer sub-threshold sweeps at $n = 10, 11$. No analytical lower-bound argument is available.

The formula $4 \cdot 3^{n-2}$ is *false* as a lower bound at $n \in \{5, \ldots, 8\}$, where the small-n absorber witnesses give $32 \cdot 3^{n-4} < 4 \cdot 3^{n-2}$, which is why the conjecture starts at $n = 9$.

Corollary 11 (Joint consequence, conditional)**.** *If Conjectures 9 and 10 both hold, then $M_n = 4 \cdot 3^{n-2}$ for $n \geq 9$, hence $M_n^{1/n} \to 3$. This would resolve a natural connected-model variant of Knuth's 1985 asymptotic questions: the* $\limsup < 3$ *variant negatively and the* $\liminf > 2$ *variant positively. It does* not*, on its own, resolve Knuth's unqualified 1985 questions, which were posed under the relaxed-connectedness convention; transporting any of the bounds of this paper to that convention is an open problem (§§ 2.2 and 8.8), and at the one case beyond $n = 4$ where both values are now known ($n = 5$, theorem 12) the two conventions agree.*

Evidence for Conjecture 10 is the subject of §§ 6 and 7.

3.4 Transport to the relaxed-connectedness convention, first data

The 1985 Stanford seminar [8] worked under a relaxation of Dijkstra's property (6) in which the legitimate states were allowed to split into several disjoint good cycles (§ 2.2). Write M_n^{rel} for the minimum state product under that convention. Every connected-valid system is also relaxed-valid, so $M_n^{\mathrm{rel}} \leq M_n$; the reverse direction is open at every $n \geq 5$ and has stood open since the 1985 formulation.

Theorem 12 ($M_5^{\mathrm{rel}} = 96$, SMT-certified)**.** $M_5^{\mathrm{rel}} = M_5 = 96$.

The proof is a z3 SMT encoding of the five validity properties in the relaxed convention: at each of the five $n = 5$ sub-threshold multisets $\{2^5\}, \{2^4, 3\}, \{2^4, 4\}, \{2^3, 3^2\}, \{2^4, 5\}$ the encoding returns UNSAT in under 90 s on consumer hardware, and an at-threshold SAT check on $\{2^3, 3, 4\}$ ($\prod m = 96 = M_5$) confirms the encoding is correct.

Encoding of property (4). Property (4) (convergence) is infinitary: "no daemon schedule starting from a bad configuration cycles back to itself." The SMT encoding reformulates it in the standard rank-function style: the solver is asked to exhibit a per-configuration rank function $\rho : \mathrm{Config}(\mathbf{m}) \to \mathrm{Fin}(B)$ with a bound B polynomial in $\prod m_i$, such that every bad-move $c \to c'$ satisfies $\rho(c') < \rho(c)$. Existence of such a ρ is a first-order condition on the $\mathrm{Fin}(m_i)$-valued rule tables and is equivalent to finite termination from every bad configuration. UNSAT at a sub-threshold multiset then means “no relaxed-valid rule table admits such a rank function,” which is the same as “no relaxed-valid rule table exists” given finite state. The soundness assumption is standard: termination on a finite transition system is witnessed by a decreasing rank, and the encoding chooses B large enough that the rank is expressive. The at-threshold SAT check extracts a rank and re-runs the extracted rule table through the Python relaxed verifier, which closes the encoding/semantics gap end-to-end.

The per-multiset UNSAT table, runtimes, and the encoding source are shipped with the repository bundle; the detailed discussion is deferred to § 8.8.

At $n = 6$ seven of the twelve sub-threshold multisets are UNSAT-certified under the same encoding, including the hardest-so-far $\{2^3, 3^3\}$ (UNSAT in ≈ 1900 s) and $\{2^4, 3, 4\}$ (UNSAT in ≈ 2170 s); the remaining five (at products $192, 224, 240, 256, 256$) return z3 UNKNOWN within a 3600 s per-multiset budget and require either a stronger WLOG symmetry break, a portfolio of solvers, or a tighter fairness encoding for certification. theorem 12 is the first case at $n \geq 5$ in which $M_n^{\mathrm{rel}} = M_n$ has been shown (at $n = 4$ the conventions disagree, $M_4^{\mathrm{rel}} \leq 16 < 24 = M_4$). Equivalently, theorem 12 resolves the question of whether $M_5^{\mathrm{rel}} < M_5$, in the negative.

4 Upper-bound constructions

This section describes the witness families supporting the conjectured bound $M_n \leq 4 \cdot 3^{n-2}$ for $n \geq 9$, with Lean certification through $n = 10$ and Python verification through $n = 18$. For $n \geq 19$ the paper falls back on Dijkstra’s classical $M_n \leq 3^n$ [4]. For $n \in \{5, \ldots, 8\}$ a distinct, tighter family supplies $M_n \leq 32 \cdot 3^{n-4}$. We begin with the unconditional base case $n = 4$ in miniature.

4.1 A fully worked example: $n = 4$

Before describing the infinite witness families we assemble the entire $n = 4$ certificate in one place. The case is unconditional (theorem 15) and illustrates in miniature every object used later.

State vector. $\mathbf{m} = (2, 2, 2, 3)$: three binary processors at positions $0, 1, 2$ and one ternary processor at position 3. Product 24.

Upper bound ($M_4 \leq 24$). A verified good cycle on $(2, 2, 2, 3)$ together with a rank-based convergence certificate (a per-configuration potential $\rho : \mathrm{Config}(\mathbf{m}) \to \mathbb{N}$ strictly decreasing on every bad-configuration move) gives $M_4 \leq 24$. The full transition table, the cycle in configuration form, and the rank potential are emitted by the parity runner of appendix A.1 and checked by `M_4_upper` in `LeanMn/SmallN/Theorem.lean`.

Lower bound ($M_4 \geq 24$). The only multiset with $\prod m_i < 24$ and each $m_i \geq 2$ on four positions is $(2, 2, 2, 2)$ at product 16. A direct case analysis on $(2, 2, 2, 2)$ in the six-property connected model eliminates every candidate good cycle: any good cycle on $\{0, 1\}^4$ would have to visit 16 distinct configurations, but the connectedness requirement (property (6)) together with mutual exclusion forces the good cycle to be a single Hamiltonian cycle on $\{0, 1\}^4$, and no rule table satisfying both closure and convergence is extensible to such a cycle. Under the relaxed-connectedness convention of the 1985 seminar the same multiset admits the historical Gray-code witness $M_4^{\mathrm{rel}} \leq 16$ [8, p. 70]; the two conventions disagree at $n = 4$ precisely because property (6) excludes the Gray-code from the connected model. The Lean proof is `M_4_lower`; the bundle is `M_4_eq_24`.

What this case illustrates. The $n = 4$ row of theorem 5 is the only one beyond $n = 3$ that is unconditional (Lean-proved end-to-end without an exhaustive-search certificate); the rows $n \in \{5, \dots, 9\}$ (theorem 5) depend on appendix C's exhaustive-search certificates. It is also the smallest instance of the relaxed-vs-connected gap studied in §§ 3.4 and 8.8. The ingredients (explicit good cycle, rank potential, case analysis eliminating sub-threshold multisets) reappear in every later construction; at $n \geq 5$ the last two scale up into the convergence rank tables of § 4.4 and the exhaustive-search certificates of appendix C respectively.

4.2 The small-n absorber family ($5 \leq n \leq 8$)

The small-n witnesses use *three binary processors*, *one quaternary processor*, and ternary fill at the remaining positions. In the stored witnesses the binary processors are in some cases clustered into a 3-consecutive block and in other cases include 2-consecutive plus a single isolated binary; the exact placement is witness-specific. The feature relevant to § 5.3 is *adjacency*: every stored witness has at least one cyclically adjacent pair of binaries, hence lies outside ARG's non-adjacency hypothesis.

Table 2 lists the placements of the four stored witnesses (file references in appendix E).

Table 2: Small-n absorber witness placements. Each row gives the state vector and the empirically measured good-configuration count L. Every row has at least one cyclically adjacent pair of binary processors.

n	$\mathbf{m}$	L	label
5	$(2, 2, 2, 3, 4)$	18	`witness_n5`
6	$(2, 2, 2, 4, 3, 3)$	35	`witness_n6`
7	$(3, 2, 2, 2, 3, 4, 3)$	52	`witness_n7`
8	$(2, 2, 3, 4, 3, 3, 2, 3)$	55	`witness_n8`

State product is $2^3 \cdot 4 \cdot 3^{n-4} = 32 \cdot 3^{n-4}$, independent of the placement subject to the validity constraint. The good cycles are mixed bounce-with-detours: neither pure sweeps nor pure bounces. No closed-form description of $L(n)$ is known in this regime, and the cycle and mover pattern change qualitatively from one n to the next.

Informally, the quaternary processor provides enough state budget to absorb the wave across the binary block; at $n = 9$ the ring is too long and a single quaternary can no longer carry the wave (consistent with the empirical fact that every orientation of $\{2^3, 4, 3^5\}$ fails at $n = 9$; see § 5.2).

Theorem 13 (Small-n upper bound, Lean-checked)**.** *`upper_bound_small` in `LeanMn/UpperBound/Theorem.lea` proves $M_n \leq 32 \cdot 3^{n-4}$ for $n \in \{5, 6, 7, 8\}$ (sorry-free; finite per-n rank-table checks discharged by `native_decide` on sealed certificates).*

4.3 The CLB ternary-strip family ($n \geq 9$, conjectured sharp)

State vector. $(2, 3, 3, \dots, 3, 2)$: binary endpoints, ternary fill. Product $4 \cdot 3^{n-2}$.

Good cycle. An up-down bounce with mover sequence $0, 1, 2, \dots, n-1, n-2, \dots, 1, 0, 1, 2, \dots$ closing at period $T_\mu = 3n - 2$. The number of good configurations is $L = n^2 - 2n + 8$.

Table 3 summarizes the cycle statistics. Python verification of the construction is via `clb_inherent_cycles.py` (see appendix F) for $n \in \{5, \dots, 18\}$, covering up to 172M configurations at the largest tested n.

Table 3: Cycle statistics of the CLB ternary-strip construction.

quantity	value
mover-pattern period T_μ	$3n-2$
good-configuration count L	n^2-2n+8
worst-case convergence steps	$\lfloor(3n^2-4n-11)/4\rfloor = \Theta(n^2)$
free transition entries fixed for liveness	$n-3$

4.4 The CUP-2 Lean-checked variant

The same ternary-strip state vector $(2, 3, \ldots, 3, 2)$ supports a second valid rule table, *CUP-2* (short for *Convergent Universal Primer, second iteration*; the name is inherited from the construction's internal label in the Lean formalization and carries no mathematical content beyond identifying the rule table), distinct from the CLB rule table of § 4.3. CUP-2 has the same mover-pattern period $T_\mu = 3n-2$; its good-configuration count is $L = (n+2)(n+3)/2 - 5$. The rule table has 87 entries that are n-independent. Each entry is tagged by up to three non-exclusive per-entry properties, namely:

- *privileged*: the entry is a move-entry (it fires on the good cycle, writing a new value different from the current center coordinate);
- *copy-neighbor*: the output equals the left or right neighbour's current value (a local-propagation entry);
- *anomalous*: the entry falls outside the two cleanest ternary-strip idiom families and required manual fit to close convergence.

A single entry may satisfy more than one property (for instance, a privileged entry that writes the value of a neighbour is both privileged and copy-neighbor); the three properties partition neither the rule table nor the privileged entries. The per-property counts across all 87 entries are 45 privileged, 40 copy-neighbor, and 5 anomalous; these sums exceed 87 precisely because the categories overlap.

CUP-2 is a distinct ternary-strip rule table with a structurally cleaner convergence certificate; its good-cycle count is comparable to, and for $n \geq 6$ strictly smaller than, the CLB good-cycle count ($L_{\text{CUP}} - L_{\text{CLB}} = -(n-4)(n-5)/2$, zero at $n \in \{4, 5\}$ and negative thereafter). The Lean formalization uses a per-n rank function on bad configurations that strictly decreases on every bad step, sealed by `native_decide` on a precomputed rank table.

Theorem 14 (CUP-2 upper bound, Lean-checked)**.** *`upper_bound_cert` in `LeanMn/UpperBound/Theorem.lean` proves $M_n \leq 4 \cdot 3^{n-2}$ for $n \in \{4, \ldots, 10\}$ (sorry-free; finite per-n rank-table checks discharged by `native_decide` on sealed certificates).*

4.5 Verification status

Table 4 consolidates the verification status across n.

Theorem 15 ($M_4 = 24$, Lean-checked)**.** *`M_4_eq_24` in `LeanMn/SmallN/Theorem.lean` proves $M_4 = 24$ in the central-daemon model (both bounds packaged; sorry-free).*

This is the Lean-level statement of the $n = 4$ row of theorem 5; theorem 15 is the artifact the paper cites when the $n = 4$ identity is invoked unconditionally (§ 4.1).

Table 4: Verification status of the upper-bound constructions in the connected model of § 2.3. "Lean" means sorry-free Lean 4 / Mathlib formalization with finite-table checks discharged by `native_decide` on sealed certificates; "Python" means `verify_system` passes on the stored witness; "loose" means ratio to $4 \cdot 3^{n-2}$ is $3^n/(4 \cdot 3^{n-2}) = 9/4$. The relaxed-connectedness $M_4 \leq 16$ Gray-code row is listed separately in table 5.

n range	construction	shape	product	status
$n = 3$	binary Gray code	$(2, 2, 2)$	8	exact in connected model
$n = 4$	special 4-ring witness	$(2, 2, 2, 3) = \{2^3, 3\}$	24	exact, Lean `M_4_eq_24`
$n \in \{5, \dots, 8\}$	small-n absorber	$\{2^3, 4, 3^{n-4}\}$	$32 \cdot 3^{n-4}$	exact†, Lean `upper_bound_small` + exhaustive LB
$n = 9$	ternary strip	$(2, 3, \dots, 3, 2)$	$4 \cdot 3^7 = 8748$	exact†, Lean UB + exhaustive LB
$n = 10$	ternary strip (CUP-2)	$(2, 3, \dots, 3, 2)$	$4 \cdot 3^8 = 26244$	conjectured sharp; Lean UB only
$n \in \{11, \dots, 18\}$	ternary strip (CLB)	$(2, 3, \dots, 3, 2)$	$4 \cdot 3^{n-2}$	conjectured sharp; Python UB only
$n \geq 19$	Dijkstra Solution 3	uniform 3-state	3^n	classical fallback, factor $\frac{3^n}{4 \cdot 3^{n-2}} = \frac{9}{4}$ above conjectured bound

† The "exact" entries for $n \in \{5, \dots, 9\}$ are certificate-backed: the Lean-checked upper-bound halves are self-contained, and the matching lower bounds are discharged by the exhaustive-search procedure of appendix C (worked-example cert streams shipped under `artifacts/rejections/`; remaining per-orientation streams regenerable from a clean clone, see appendices C.10 and C.11). The $n \in \{3, 4\}$ rows are unconditional today.

Table 5: Historical comparison: constructions that are valid only under the relaxed-connectedness convention of the 1985 seminar [8], not in the connected model of this paper. Listed here for provenance; *not* witnesses for M_n as defined in § 2.5.

n	construction	shape	product	convention
$n = 4$	binary Gray code	$(2, 2, 2, 2)$	16	relaxed (disjoint 8-cycles)

4.6 A historical looser construction

The construction $\mathbf{m} = (2, 3, 3, \dots, 3)$ (one binary at an endpoint, ternary elsewhere) gives $M_n \leq 2 \cdot 3^{n-1}$ for $n \geq 3$. This is looser than $4 \cdot 3^{n-2}$ by a factor 3/2 and is recorded for historical interest only; it is the simplest valid non-uniform family beyond Dijkstra's originals.

5 Exact lower bounds at small n

This section certifies the $\geq$ direction of theorem 5. The methodology and the per-n enumeration are described here in summary; the full reproducibility material lives in appendix C.

5.1 Methodology (certificate-grade)

For each $n \in \{3, \dots, 9\}$ we proceed as follows.

1. Enumerate all state-count vectors with product T satisfying $T < M_n$ (strictly below the claimed minimum; candidates at $T = M_n$ are excluded so the actual sharp witness is not

rejected).

2. Quotient by cyclic rotation, reflection, and state-renaming symmetries where sound.

3. Enumerate candidate good cycles and partial rule tables within each class.

4. Apply sound pruning rules (quasi-unidirectionality, forced-neighbor constraints, detOf-dictionary consistency).

5. For every surviving candidate, verify that no completion to a full transition function satisfies the six validity properties of § 2.3.

6. Produce machine-checkable failure certificates or independently replayable logs.

7. Validate the search with an independent checker (a re-implementation in a second code path, appendix C).

Remark 16. The $\geq$ direction of theorem 5 is *not* merely the statement that the search failed to find a witness; it is that every sub-target product class is eliminated by a sound, replayable certificate. Appendix C is the proof obligation for $n \in \{5, \dots, 9\}$, discharged by the deterministic exhaustive-search driver; the pre-generated rejection stream at `artifacts/rejections/` ships in full for $n \in \{3, 4\}$ and for the all-binary orientation at $n = 5$, with per-cert streams at every other $(n, \text{orientation})$ pair regenerable from a clean clone (appendix C.11, appendix C.10).

5.2 The exhaustive verification at $n \in \{3, \dots, 9\}$

All state-count vectors with $\prod_i m_i < M_n$ are eliminated by the above method. In particular at $n = 9$:

- every multiset with $\prod_i m_i = 7776 = 32 \cdot 3^5$ fails: the three multisets $\{2^3, 3^5, 4\}$, $\{2^4, 3^4, 6\}$, and $\{2^5, 3^3, 9\}$ (appendix C, table 10, lists every such multiset and its dihedral orientation count; total 94 orientations);

- every multiset with $\prod_i m_i \in (7776,\ 8748)$ fails: appendix C, table 11, lists all 24 such multisets with their products;

- the state vector $(2, 3, 3, 3, 3, 3, 3, 3, 2)$ at product 8748 is a verified valid system (CLB witness, § 4.3; also CUP-2 witness, § 4.4).

The three 7776 multisets and the 24 multisets in $(7776, 8748)$ were discharged by targeted computational probes on these absorber-specific candidates rather than by running the general App C rejection stream end-to-end (the latter is feasible in principle at $n = 9$ on local hardware but the per-multiset per-orientation wall-clock and output-size cost make pre-generation of the full bundle impractical; appendix C.10). The probes themselves are deterministic Python scripts shipped under `docs/axisc/` and `probes/`; their outputs (per-orientation verdicts and Farkas-style infeasibility witnesses where available) are archived as JSON alongside the source. Hence $M_9 = 8748$ exactly. The same App C methodology, applied at the smaller target products $M_n = 32 \cdot 3^{n-4}$ for $n \in \{5, 6, 7, 8\}$ and $M_n = 24$ for $n = 4$ and $M_n = 8$ for $n = 3$, certifies the remaining rows of theorem 5 (with the $n = 4$ row also discharged analytically in Lean as theorem 15).

State-space size grows as $\prod m$, and the number of candidate good cycles, partial rule tables, and completions grows substantially faster than that. Beyond $n = 9$ the certificate search exceeds available hardware budget under the current pruning rules; this is why Conjecture 10 is open for $n \geq 10$ (§ 5.4).

5.3 ARG's bound and the $n = 9$ crossover

The $n = 9$ ratio crossover is *not* explained by ARG's 1985 LCM bound. Two observations suffice [8, §2–§3]; a self-contained dispatch is shipped with the repository bundle (appendix C.11).

1. **Scope.** The small-n absorber witnesses use adjacent binary positions (§ 4.2), which lie outside ARG's non-adjacency hypothesis. ARG's bound is silent on every known valid construction in this family.

2. **Insensitivity.** On the non-adjacent orientations of $\{2^3, 4, 3^{n-4}\}$ that ARG does cover, the LCM functional is constant or strictly nested between $n = 8$ and $n = 9$ under each of four natural readings of "LCM of block state counts" (global, per-arc-min, per-arc-max, block-product). A shipped script (`arg_lcm_finite_checks.py`; appendix F) enumerates the dihedral orbit of non-adjacent orientations (6 at $n = 8$, 10 at $n = 9$) and reports for each reading the set of values taken across the orbit; the verified numerics are *global-LCM* $\{12\} = \{12\}$, *per-arc-min-LCM* $\{3, 12\} = \{3, 12\}$, *per-arc-max-LCM* $\{12\} = \{12\}$, and *block-products* $\{3, 4, 9, 12, 27, 36\} \subset \{3, 4, 9, 12, 27, 36, 81, 108\}$ (first set at $n = 8$, second at $n = 9$). No LCM-functional reading discriminates the phase-transition boundary.

Remark 17 (ARG's bound remains a genuine 1985 contribution)**.** Within its stated non-adjacency hypothesis ARG's result is correct and useful: it reproves all-binary impossibility for $n > 4$ ($L = 1$ forces $N = 0$) and caps the binary count at 4 in the all-ternary-non-binary mixed case $\{2^k, 3^{n-k}\}$ ($L = 3$ forces $k \leq 4$). These are scope-limits on the witness search; they simply do not locate the $n = 9$ boundary.

The 1985 seminar's quasi-unidirectionality bound (no 4 or more consecutive binary processors [8, p. 75]) is compatible with the crossover but silent on the 3-consecutive case at $n \geq 9$. Candidate analytical directions for future work include palindromic entry conflict, shadow-cycle mirror arguments, and good-cycle-length versus product entropy bounds; we treat the mechanism as an open structural problem.

5.4 Why exhaustive search does not extend to $n \geq 10$

At $n = 10$, the target state space itself has size 26244, but exhaustive lower-bound certification requires eliminating many state-count vectors, candidate good cycles, partial rule tables, and completions below the target. Under the current pruning rules, that certificate search exceeds the available budget. appendix C reports the measured growth in candidate-count and wall-clock from $n = 5$ through $n = 9$, from which the $n = 10$ extrapolation follows.

6 The circulation detector and structural evidence

6.1 The lifted-defect graph

Fix a candidate good cycle gc on $\mathbf{m}$. The *lifted-defect graph* has vertex set

$$V_{\text{lift}}(gc) = \{(k, q, a) : k \in \text{Fin}(L),\ q \in \text{Fin}(n),\ a \neq c_k(q),\ c_k[q := a] \in T_{N_1}(gc)\}$$

with projection $\pi(k, q, a) = c_k[q := a]$. Edges are induced by the firings of the cycle and classified into four types according to the relative position of the cycle mover at step k and the defect coordinate q:

- **transport**: the mover at step k is far from q (at distance ≥ 2);
- **c_self**: the mover at step k is exactly q;
- **c_left**: the mover is at position $q - 1$;

- **c_right**: the mover is at position $q + 1$.

The 4-type classification partitions the forward edges of the lifted graph (verified by the `classify_type` audit on 4795 forward edges; see appendix B). The 3-type collapse (transport, c_sided := c_left ∪ c_right, c_self) is direction-covariant.

6.2 The circulation detector

For a given good cycle gc, set up the linear program

$$\text{find } \Phi : E_{\text{lift}} \to \mathbb{Q}_{\geq 0} \quad \text{with} \quad B^\top \Phi = 0 \quad \text{and} \quad \textstyle\sum_e \Phi_e = 1,$$

where B is the incidence matrix of the lifted-defect graph. The positive-mass normalization $\sum_e \Phi_e = 1$ replaces the abstract $\Phi \neq 0$ condition with a bona-fide linear constraint, so that the search is an actual linear program rather than a feasibility predicate. By Hoffman's circulation theorem [10], this LP is feasible if and only if the lifted graph contains a directed cycle. This step is a theorem.

Remark 18 (Detector status)**.** The detector *predicts*, empirically, that circulation feasibility corresponds to obstruction: on every tested record, feasible circulation appears exactly at sub-threshold candidates and infeasible circulation appears exactly at valid-system witnesses (sharp or super-threshold). A proof would require showing that a lifted directed cycle forces peelTube(gc) to be nonempty. The downstream consumers `sk_nonempty_via_tube_{small,large}_n` turn peelTube(gc).Nonempty into SK(gc).Nonempty, and `SinkKernel.not_converges_of_SK_nonempty` turns SK(gc).Nonempty into nonconvergence; these two steps are established. The missing step is "lifted circulation ⇒ peelTube(gc).Nonempty."

6.3 Empirical results: the 97-record stratified corpus

Three terminology labels (from definition 4) are used throughout this and the next subsection: *sub-threshold candidate record*, B_n*-threshold valid witness*, and *valid super-threshold witness*.

Table 6 gives the corpus composition.

Table 6: The extended stratified corpus. No record is duplicated; small-n absorber and ternary-strip witnesses use distinct state vectors at every n in the corpus. CUP-2 at $n = 4$ is excluded. The $n = 8$ sub-threshold row and the $n = 9$ Table 7 row were added under the strengthening effort documented in the supplementary memo listed in appendix F; the four $n = 9$ LP-infeasible sub-threshold records are discussed in § 6.4. The canonical 97-record artifact carrying LP and Axis C verdicts for every row is `corpus_canonical.json` (overall hash `f4b017b1f57687cc`; appendix F).

class	n values	count	LP status
sub-threshold candidates	$n \in \{5, 6, 7\}$	19	feasible (19/19)
sub-threshold candidates (full sweep)	$n = 8$	59	feasible (59/59)
sub-threshold candidates (Table 7)	$n = 9$, $\prod m = 7776$	9	feasible (5/9); see §6.4
small-n absorber (exact-sharp at n)	$n \in \{5, 6, 7, 8\}$	4	infeasible (4/4)
ternary-strip (CLB / CUP-2)	$n \in \{5, 6, 7, 8, 9, 10\}$	6	infeasible (6/6)
total		97	83 feasible + 14 infeasible

Notes on the corpus. The 10 infeasible records are valid-system witnesses; among them, the 4 small-n absorbers are sharp at their n (theorem 5) and the 6 ternary-strip records are sharp only at $n = 9$ among the currently stated exact values. The ternary-strip records at $n \in \{5, \ldots, 8\}$ are valid super-threshold witnesses (strictly above the small-n absorber's tighter

M_n); the ternary-strip record at $n = 10$ is conjecturally sharp. The four small-n absorbers and the six ternary-strip witnesses use distinct state vectors at every n (e.g. $(2, 2, 2, 3, 4)$ versus $(2, 3, 3, 3, 2)$ at $n = 5$); no de-duplication is required.

Stratification rule (independent of LP outcome). The sub-threshold candidates are drawn by a deterministic enumeration over composition classes on $\mathbf{m}$ (sorted multisets with $\prod m_i < B_n$), with one representative per class passed to candidate-good-cycle enumeration. The selection rule is defined in terms of the composition class and the driver seed only; it is *not* a function of the LP verdict or of any quantity computed downstream of the LP. This matters for the separation claim: the corpus would look the same under any alternative detector, and the LP was not used to reject or select records.

Sub-threshold corpus coverage. The 19 core sub-threshold candidates are drawn at $n \in \{5, 6, 7\}$. A strengthening-effort extension (§ 6.4; see the supplementary memo listed in appendix F) runs the same enumeration driver over the full set of 59 unordered multisets with product $< B_8 = 2592$ at $n = 8$, and over the three product-7776 Table 7 multisets at $n = 9$; cycles are found on every $n = 8$ multiset and on 9 (multiset, ordering) pairs at $n = 9$. At $n = 10$ no sub-threshold enumeration was attempted.

Result. On $n \in \{5, 6, 7, 8\}$ the circulation-LP detector separates every record in the stratified corpus: all $19 + 59 = 78$ sub-threshold candidate records at those n are circulation-feasible, and all 4 small-n absorber witnesses and 4 ternary-strip witnesses (at $n \le 8$) are circulation-infeasible. At $n = 9$ the sub-threshold side fractures under C1: 5 of 9 tested Table 7 sub-threshold orderings are LP-feasible, and 4 are LP-infeasible; the infeasible sub-family is documented in § 6.4, and a strict upgrade of the detector (Axis C, § 6.5) recovers all four while preserving separation on every other tested record.

6.4 Detector failures at $n = 9$

Extending the sub-threshold enumeration to the three product-7776 multisets of Table 7 (§3.2: $\{2^3, 3^5, 4\}$, $\{2^4, 3^4, 6\}$, $\{2^5, 3^3, 9\}$) surfaces a concrete localized failure of the C1 circulation-LP detector. On $\{2^3, 3^5, 4\}$ the enumeration finds candidate good cycles on five cyclic orderings (up to dihedral equivalence); four of these orderings return LP-infeasible. All four share the same structural signature:

- $k_{\text{bin}} = 3$ (three binary processors);
- `bin_adj_pairs` $= 0$ (no two binary positions are cyclically adjacent);
- min-binary-gap $= 2$ (binary positions are pairwise separated by at least one ternary and nowhere consecutive).

The four orderings are, with cycle length L:

ordering	L
$(2, 3, 2, 3, 3, 3, 2, 3, 4)$	47
$(2, 3, 2, 3, 3, 3, 2, 4, 3)$	47
$(2, 3, 2, 3, 2, 3, 3, 3, 4)$	47
$(2, 3, 2, 3, 3, 3, 3, 2, 4)$	44

On each record the LP returns optimum exactly 0 with empty support, the lifted-defect digraph on the Hamming-1 tube admits no non-trivial non-negative circulation. The remaining fifth ordering of $\{2^3, 3^5, 4\}$ has `bin_adj_pairs` $= 1$ and is LP-feasible. On $\{2^4, 3^4, 6\}$ all four tested orderings are LP-feasible, including $(2, 3, 2, 3, 2, 3, 2, 3, 6)$ whose binaries are pairwise non-adjacent; the failure locus is therefore not *non-adjacent binaries in general* but specifically the intersection with $k_{\text{bin}} = 3$.

Coverage of the third product-7776 multiset. $\{2^5, 3^3, 9\}$ is listed alongside the other two product-7776 $n = 9$ multisets in table 10 and has 28 dihedral orientations, but contributes zero cycles to the detector corpus. The candidate-good-cycle enumerator finds no good cycle on any of its orderings within the probe budget used for table 6; the multiset is therefore eliminated cycle-free from the detector side (there is nothing for C1 or Axis C to fire on) and is separately ruled out, as a sub-threshold product class, by the exhaustive search of § 5.2. No $\{2^5, 3^3, 9\}$ record was silently dropped from the corpus for a verdict-dependent reason.

Interpretation. The four counterexamples are sub-threshold records (the multiset $\{2^3, 3^5, 4\}$ at $\prod m = 7776 < 4 \cdot 3^7 = M_9$; the product is sharp-excluded from valid by the RC and AC exhaustive sweeps cited in § 5.4). The C1 detector failing to fire, i.e. returning LP-infeasible, on them is a detector failure, not a validity claim about the multiset. Importantly, the structural sub-family on which the detector fails coincides with the *non-adjacent binary* zone that ARG's 1985 LCM bound treats (§5.3), and the same sub-family on which ARG's LCM-functional is constant between $n = 8$ and $n = 9$. The C1 LP is silent on the same sub-family as ARG's bound: it is a detector on every tested record *outside* the non-adjacent $k = 3$ binary zone and is silent inside it.

Restricted-LP variants. The T+sided-only variant (§6.7, the $c_{\text{self}} = 0$ restriction of C1) does *not* recover the counterexamples: all four records remain LP-infeasible under T+sided-only with identical objective 0, support size 0, and the same per-record edge statistics. The transport-only variant is trivially infeasible on all nine $n = 9$ records. Concretely, the design-space for an $n \geq 9$ detector that handles this sub-family is catalogued in the supplementary memo listed in appendix F, where a restricted-to-edge-type construction is marked RED and two alternative axes (full-forced-NG sink-kernel detector, structural template-match against the ternary-strip witness) are open.

Relation to existing open question. The restricted- feasibility fact of §6.7 (*c_self edges are empirically unnecessary on every tested* $n \leq 8$ *sub-threshold record*) remains intact: on the five LP-feasible $n = 9$ records the T+sided-only LP preserves feasibility with unchanged support size, and on the four counterexamples both C1 and T+sided-only are infeasible. The open question stated in Conjecture 21 and § 8.1 updates to: *every good cycle at* $n \geq 9$ *with product below* $4 \cdot 3^{n-2}$ and *outside the* $k_{\text{bin}} = 3$ *non-adjacent-binary sub-family has a directed cycle in the lifted transport-plus-sided graph; the sub-family is conjectured to require a distinct LB mechanism.*

6.5 The Axis C detector for $n \geq 9$

The four $n = 9$ counterexamples of § 6.4 are recovered by a strict upgrade of the circulation detector that leaves the Hamming-1 tube: the *sink kernel of the candidate's forced-NG graph*. The upgrade is machine-rechecked on the full 97-record canonical corpus artifact (overall hash `f4b017b1f57687cc`; archived with the repository bundle of appendix C.11); the 72-record slice previously shipped covers 4 at_smallN absorbers + 59 $n = 8$ sub-threshold + 9 $n = 9$ Table-7 sub-threshold records, with the remaining 25 (19 sub-threshold at $n \in \{5, 6, 7\}$ + 6 at_clb witnesses at $n \in \{5, \ldots, 10\}$) filled in by the reconciliation pass documented in the supplementary memo listed in appendix F. The memo consolidating all eight probe axes is also listed in appendix F.

Definition. Given a candidate record $(\mathbf{m}, \text{cycle}, \text{movers}, \text{detOf})$ let $\text{NG} = \prod_i \mathbb{Z}_{m_i} \setminus \text{cycle}$ be the non-good configurations and let the *forced-NG digraph* on NG have an edge $c \to c'$ whenever some processor p has $\text{detOf}[(p, c_{p-1}, c_p, c_{p+1})] \neq c_p$ and applying that move yields $c' \in \text{NG}$. The *sink kernel* is the largest $S \subseteq \text{NG}$ such that every $c \in S$ has at least one out-edge into S (equivalently, the fixed point of iterative removal of vertices with no out-neighbor in the current set). The Axis C detector *fires* on a record iff its sink kernel is nonempty.

Monotonicity certificate. SK is defined by a coclosure-style fixed point, it is the largest subset S of NG in which every vertex has at least one out-neighbor in S. Adding forced edges can only add out-neighbors to a given vertex; it cannot remove any. Consequently, every vertex that qualified for membership in S under the smaller edge set still qualifies under the larger one, and SK grows monotonically in the edge set (equivalently, in detOf): adding forced edges can only enlarge SK. Every extension of the partial detOf to a full transition function f adds forced edges, so $\mathrm{SK}(f) \supseteq \mathrm{SK}(\mathrm{detOf})$. A valid self-stabilizing system has $\mathrm{SK}(f) = \emptyset$ under its full f (convergence forbids any closed set of non-good configs under forced moves). Hence: $\mathrm{SK}(\mathrm{detOf}) \neq \emptyset$ *on a candidate record* $\Rightarrow$ *no extension of* detOf *to a valid system exists.* This is the one-sided sub-threshold certificate the circulation LP provides for $n \leq 8$, now extended past the $n = 9$ sub-family on which C1 was silent.

Corpus verdict. Axis C fires on all 87 sub-threshold records of the 97-record canonical corpus (including all 4 $n = 9$ counterexamples of § 6.4) and is silent on all 10 valid witnesses (4 small-n absorbers at $n \in \{5, \dots, 8\}$ plus 6 at_clb ternary-strip witnesses at $n \in \{5, \dots, 10\}$): separation is 97/97 overall. The 87 sub-threshold count decomposes as 19 records at $n \in \{5, 6, 7\}$, 59 at $n = 8$, and 9 at $n = 9$ (Table 7); the 68-record sub-threshold slice sometimes cited separately is the $59 + 9$ portion at $n \in \{8, 9\}$. On the four $n = 9$ counterexamples $|\mathrm{SK}|/|\mathrm{NG}|$ lies in $[0.26, 0.50]$ with nontrivial strongly-connected-component counts $\{12, 16, 38, 12\}$ and largest-SCC sizes $\{1020, 1050, 1185, 930\}$; the peel geometry is substantial and is the concrete object targeted by the analytical open problem below.

Cross-check: the verdict is genuine, not a det-coverage artifact. Before adopting Axis C we audited it against a potential artifact: the small-n absorber records carry a full detOf dictionary while the $n = 9$ sub-threshold records carry only cycle-visited triples (typical coverage $\sim 0.6\%$). The concern was that $|\mathrm{SK}| = 0$ on the absorber records might merely reflect full-coverage forcing, not a structural property. Stripping each absorber's detOf down to exactly the cycle-visited triples and recomputing gives $|\mathrm{SK}| = 0$ on all four (with 168, 480, 1812, 6876 stripped triples respectively). The signal is therefore structural, not coverage-driven; the sibling axis D (identity-completion verifier) fires 4/4 but was traced to the coverage asymmetry and is retired.

Relation to the circulation LP. The C1 circulation LP of § 6.2 is a strictly weaker detector: on the 97-record canonical corpus, C1 feasible $\Rightarrow$ Axis C fires holds with zero counterexamples, and the four $n = 9$ records in Axis C $\setminus$ C1 are exactly the non-adjacent $\{2^3, 3^5, 4\}$ orderings. Structurally: a feasible C1 gives a non-trivial non-negative circulation on the Hamming-1 tube, and that circulation lifts to a directed cycle in the forced-NG graph at Hamming distance ≤ 1 from the good cycle; Axis C allows forced cycles at any Hamming distance, which is where the four $n = 9$ counterexamples live.

Analytical status. Axis C's detector verdict on the corpus is machine-recheckable at $O(|\mathrm{NG}|)$ per record (no LP solve); the structural certificate (sink-kernel monotonicity) is proved. What remains open is the *uniform* statement: whether $\mathrm{SK}(\mathrm{detOf}) \neq \emptyset$ holds on *every* sub-threshold candidate record at arbitrary n, not just the 87 sub-threshold records in the canonical corpus. This is a cleaner target than the C1 "feasible $\Rightarrow$ peelTube-nonempty" implication: Axis C is not restricted to the Hamming-1 tube, and the sink-kernel is a discrete topological invariant of $X(\mathbf{m}) \setminus$ cycle (a closure property of the forced-move digraph, not an arithmetic one on detOf). It is the "topological invariant forbids it"-shaped obligation the analytical program has been trying to isolate.

Extended $n = 10$ sub-threshold sweep. The 87 sub-threshold records of the canonical corpus above cover $n \in \{5, \ldots, 9\}$; $n = 10$ is absent because the small-n probe budget used for the canonical corpus does not reach $n = 10$ cycle enumeration. A separate full-$n = 10$ sweep probes every one of the 291 sorted ("canonical") sub-threshold $n = 10$ multisets using two orthogonal tools: (a) a DFS good-cycle enumerator (`probe_axisc_n10_sweep.py`, 30 s budget per canonical ordering, 2 orderings per multiset) and (b) a z3 SMT cycle-existence encoder (`probe_cycle_smt.py` / `sweep_smt_n10_uncovered.py`, a first-order encoding of state$[t, p]$ and mover$[t]$ variables with transition, fairness, closure, and det-consistency constraints) run on every multiset the DFS enumerator could not find a cycle for within its budget. The SMT encoding's cost is polynomial in the cycle length and independent of $\prod m$, so it closes multisets on which brute-force DFS exceeds its 30 s cap. Combining the two tools:

- DFS found good cycles on 77 multisets and ran Axis C on every cycle;
- SMT found good cycles on the remaining 214 multisets and ran Axis C on every cycle;
- across **320** total (multiset, ordering) probes covering every one of the 291 sub-threshold multisets, *Axis C fires on every probe, zero silent* (**320/320**).

Sink-kernel geometry across the probes: $|\mathrm{SK}|/|\mathrm{NG}|$ in $[0.038, 1.000]$, mean 0.175; good-cycle length L in $[20, 59]$.

Extended $n = 11$ sub-threshold sweep. The same two-tool pipeline extends to $n = 11$: the DFS enumerator on all 564 sorted sub-threshold multisets (same 30 s-per-ordering budget, 2 orderings per multiset) surfaces good cycles on 90 multisets across 112 probes, and the SMT gap-closer (`sweep_smt_n10_uncovered.py` reused with `-L-max 30`, 90 s timeout per multiset) closes 473 of the remaining 474 budget-limited multisets. Combining:

- DFS found good cycles on 90 multisets and ran Axis C on every cycle;
- SMT found good cycles on 473 of the 474 DFS-unreached multisets and ran Axis C on every cycle;
- 1 multiset ($[2^8, 4, 4, 6]$) returned UNKNOWN at the SMT timeout — a solver-budget outcome, not evidence of an obstruction;
- across **585** total (multiset, ordering) probes covering 563 of the 564 sub-threshold multisets, *Axis C fires on every probe, zero silent* (**585/585**).

Sink-kernel geometry across the probes: $|\mathrm{SK}|/|\mathrm{NG}|$ in $[0.026, 0.999]$, mean 0.110; good-cycle length L in $[22, 65]$. This extends the $n = 10$ verdict one layer further and is cleanly consistent with the uniform claim Conjectures 9 and 10 predicts at $n = 11$. Artifacts: `axc_n11_sweep_results.json` (DFS sweep), `axc_n11_smt_sweep_results.json` (SMT-closed sweep).

Scope caveat. Axis C monotonicity eliminates a specific (cycle, detOf) candidate, not a whole multiset: a multiset is ruled out only up to the (cycle, ordering, detOf) probes actually surfaced by the pipeline. The DFS half uses a fixed 30 s budget and two canonical orderings per multiset; the SMT half finds *a* good cycle per multiset, not all of them. "320/320" at $n = 10$ and "585/585" at $n = 11$ therefore mean *every probe that the two-tool pipeline surfaced fires Axis C*, not *every candidate good cycle on every multiset does.* Completeness of the good-cycle enumerator under the search's pruning discipline is the additional gap between a per-probe statement and a per-multiset statement; it coincides with the exhaustive-search completeness question of appendix C and is not separately claimed here. This is the sense in which Conjecture 10 is "Axis C-supported" at $n = 10, 11$: supported per probe, not proved per multiset. The verdict is nevertheless cleanly consistent with the uniform claim Conjectures 9 and 10 predicts at $n = 10, 11$. The $n = 10$ DFS sweep (291 multisets in 3.7 h), its SMT-closed sweep (214 budget-limited multisets, 2.8 h), the $n = 11$ DFS sweep (564 multisets), its SMT-closed sweep (473 of 474 budget-limited multisets), and the combined findings memo are all archived in the repository bundle (appendix C.11) and indexed in appendix F.

6.6 Supplementary probes

Sharpness probe. At $n \in \{5, 6\}$, feasibility transitions sharply at $\prod m = M_n$: every immediately-sub-threshold candidate is feasible and every M_n-threshold valid witness is infeasible. At $n = 7$ sub-side cycle enumeration timed out; this caveat is noted.

Random-multiset probe. 15 random sub-threshold multisets at $n \in \{5, 6, 7\}$; for the 12 that admitted a candidate good cycle, all 12 were LP-feasible. The detector generalizes beyond the `enumerate_multisets` stratification.

Perturbation probe. 11 edge-definition and orientation variations were tested; no record flipped verdict. The route is robust to edge-definition perturbation.

6.7 The restricted-feasibility observation

Primary observation (empirical). *After forbidding c_self edges entirely, circulation remains feasible on all* 19 *sub-threshold records.* This is the central observation of the subsection.

Secondary observation (selected LP optimum). On the LP optima returned by the default solver, c_self carries zero flow on all 19 records. This is a statement about a selected optimum; if the LP optimum is non-unique, some alternate optimum may place positive flow on c_self. The restricted-feasibility statement above does not depend on optimum selection.

Edge-type flow statistics (selected optima). On the orientation-invariant 3-type collapse, the selected-optimum mean support is transport 66.5%, c_sided 33.5%, c_self 0% (the split of c_sided into c_right 33.2% and c_left 0.3% is a cycle-orientation artifact: reversing the cycle swaps the two, so only the sum is a property of the record). The c_self edges constitute up to 14.9% of the lifted graph but contribute 0% in the selected optima.

On the tested corpus, the c_self edges are empirically unnecessary for circulation; forbidding them preserves feasibility on every sub-threshold record tested. This suggests that any eventual proof may be able to work entirely in the transport-plus-sided subgraph.

Analytical status. The proof attempt, a q-tube forward walk in the transport-plus-sided subgraph, stalls at a specific named obstruction. The walk advances cleanly on transport and sided-adjacent steps but hits a "hole" at every step where the cycle mover equals the defect coordinate ($\mathrm{mov}_k = q$). Every position q fires at least once per good cycle (fairness), so every q-tube walk has at least one hole. Closing the hole requires cycle-local $\mathrm{detOf}(gc)$ information. This obstruction is one of the five families catalogued in § 7.

6.8 What this section establishes (and what it does not)

§ 6 establishes:

- separation of all 97 records of the extended corpus at $n \in \{5, \ldots, 10\}$ under the Axis C sink-kernel detector (§ 6.5), and separation of 93 of 97 under the circulation-LP detector of § 6.2, the four C1-infeasible $n = 9$ sub-threshold records of § 6.4 being exactly Axis C $\setminus$ C1;
- robustness of C1 to edge-definition perturbation and random sub-threshold sampling;
- restricted feasibility without c_self on every tested sub-threshold record;

- a one-sided sub-threshold certificate via sink-kernel monotonicity on the forced-NG digraph (§ 6.5);
- a concrete obstruction mechanism pointing at the $(C, \mu) \times \mathrm{detOf}$ interaction.

It *does not* establish:

- any extension of the corpus-separation claim beyond the 97 records or beyond $n \leq 10$;
- a uniform-in-n proof that $\mathrm{SK}(\mathrm{detOf}) \neq \emptyset$ on every sub-threshold candidate (§ 6.5), nor a proof of the analogous C1 implication "feasible $\Rightarrow$ peelTube-nonempty."

The specific statement is Conjecture 21 of § 8.1; empirical support is substantial but analytical attempts fail at mover-equals-defect steps (§ 7.2).

7 A map of the obstruction landscape

This section catalogs systematic analytical attempts at the lower bound, organized by the underlying invariant family. Five families were tested. Each subsection follows the same five-point template: proposed invariant, target lemma, what was tested, failure mode, what this teaches a future attempt. Long-form descriptions of each tested approach are deferred to appendix D.

7.1 Framing

The five families are: flow-theoretic decomposition; ambient topological invariants; map-level invariants; sheaf cohomology; and arithmetic extraction. We treat each in turn.

7.2 Family 1: flow-theoretic decomposition

Proposed invariant. A per-vertex balance identity on the 4-type decomposition (transport, c_self, c_left, c_right), or in the direction-covariant restatement on the 3-type decomposition (transport, c_sided, c_self).

Target lemma. For every good cycle at $n \geq 9$ with product below $4 \cdot 3^{n-2}$ and every defect tube indexed by q, transport-plus-sided edges close a directed walk in the lifted graph.

What was tested. The 4-type per-vertex balance identity [8] (failed, 4.9% leading-order residual against a 1% discrimination target); the direction-covariant 3-type balance (failed, 2.7% residual); two analytical proof attempts of T+sided-only directed-cycle existence (§ 6.7 and appendix D).

Failure mode. The q-tube forward walk at a lifted vertex (k, q, a) advances cleanly on transport and sided-adjacent steps but hits a *mover-equals-defect hole* at every step where $\mathrm{mov}_k = q$. Fairness guarantees every q encounters such a step. Closing the hole requires cycle-local $\mathrm{detOf}(gc)$ information, preventing a uniform-in-n argument. Two independent attempts identify the same wall.

What this teaches. The obstruction is localized at mover-equals-defect steps. Progress requires either (a) a structural handle on the move-entry context at these steps that does not case-split, or (b) a proof of T+sided cycle existence that bypasses the walk construction entirely (§ 8.1).

7.3 Family 2: ambient topological invariants

Proposed invariants. π_1 and H_1 of $\mathrm{NG}(C)$; the linking matrix of mover-class subcycles; Lefschetz numbers and Artin–Mazur ζ_f; Cheeger and spectral-gap of the forced neighborhood graph; Forman–Ricci curvature; discrete Morse theory.

What they would need to prove. A homotopy- or curvature-level invariant that distinguishes sub-threshold good cycles from valid-system good cycles.

What was tested. See appendix D.

Failure mode. All tested invariants scale with ambient dimension (n, L) rather than with per-m_i information. $\pi_1 = 0$ on tested records; the linking matrix is circulant, forcing $\det \Lambda = 0$; Lefschetz is trivial by contractibility; Cheeger $\lambda_2 = 0$ (graph disconnected in both classes); Forman–Ricci is monotone in $|V|$; the Conley-style $\beta_1/|N|$ ranges $[1.087, 1.500]$ at sub-threshold and $[1.080, 1.500]$ at threshold fully overlap. The homotopy type of $\mathrm{NG}(C) \subseteq \prod \Delta^{m_i-1}$ is (n, L)-parametrized; $\mathbf{m}$ enters only via ambient-cell counts.

What this teaches. Ambient-space topology is (n, L)-parametrized. Any future invariant must see $\mathbf{m}$ beyond ambient dimension, and must address the *coverage-inversion* diagnostic: $|\mathrm{detOf}|/T_{\mathrm{total}} \approx L/(n \cdot \prod m)$ *decreases* as the product grows, pushing any monotone-in-coverage invariant in the wrong direction for a lower bound.

7.4 Family 3: map-level invariants

Proposed invariant. The Conley index of the good cycle as an isolated invariant set of the transition map.

Target. An intrinsic dynamical invariant of the system that distinguishes sub-threshold from valid.

What was tested. Scale-normalized $(\beta_0, \beta_1)/|N|$ pairs of (N, L) where $N = C \cup N_1(C)$ and L is the exit set.

Failure mode. The Conley index is homotopy-invariant under f-homotopies, but at sub-threshold no valid f exists. Any completion of the rule table (e.g. stay-completion) is arbitrary; the computed index reflects completion choice, not intrinsic dynamics. Normalized ranges fully overlap valid-system records.

What this teaches. Standard dynamical-systems invariants that assume the self-map exists cannot directly transfer. A map-level approach would need to treat the "no valid f" case explicitly, a pre-dynamical analog, a partial-function Conley framework, or a relation-valued model (§ 8.3).

7.5 Family 4: sheaf cohomology

Proposed invariant. Čech H^1 of an extension-problem sheaf on the 3-cells of $\prod \Delta^{m_i-1}$.

Target. $H^1 \neq 0$ at every sub-threshold good cycle and $H^1 = 0$ at every valid-system good cycle.

What was tested. The standard sheaf (with restriction maps the natural projection); the single-priv-propagation sheaf; the path / cycle-space sheaf on the lifted forced-neighborhood 1-skeleton; the convergence-depth-parameterized sheaf.

Failure mode. The standard sheaf's restriction maps on shared 3-cells are automatic, reducing Čech cohomology to the nerve cohomology with constant coefficients; empirically the nerve β_1 ranges over [27, 11179] at valid-system records, so $H^1 \neq 0$ fails to vanish where the conjectured separator would require it to. The three non-standard variants either compute $H^1 = 0$ uniformly, have overlapping ranges across the two classes, or invert the signal.

What this teaches. The obstruction is global (requires convergence information), but local-cellular sheaves are by construction local. A sheaf formulation would need restriction maps that encode global structure, e.g. convergence-propagation or single-priv-violation-propagation that is genuinely non-local (§ 8.4).

7.6 Family 5: arithmetic extraction

Proposed invariant. A scalar arithmetic inequality on **m**-features that separates sub-threshold from valid.

Target. Provable independence from $\prod m < B_n$, i.e. a non-tautological separator.

What was tested. A 6-feature linear regression ($R^2 = 0.809$, threshold accuracy 1.0 on 25 records); a 10-feature extension ($R^2 = 0.789$, in-sample accuracy 0.966, leave-one-out accuracy 0.931).

Failure mode. The best approximation is $\log B_n(n) - \log \prod m$, the threshold condition itself. The extraction is tautological. An additional "$n_{\text{bin}} \geq 3$ is the only separator" reading is falsified by the decisive test on the small-n absorber witnesses, which are themselves 3-binary and infeasible.

What this teaches. A viable arithmetic inequality must be provably independent of $\prod m < B_n$; the regression feature spaces tested contain no such feature. Future arithmetic attempts must pre-commit a Newness check against the problem's own definition.

7.7 Unifying observation

Observation 19. The failed approaches agree on one point: the obstruction is not visible in the ambient state space alone. It lives in the triple interaction among the good cycle C, the mover sequence μ, and the determined partial rule table $\mathrm{detOf}(gc)$.

Each family fails for the same reason from a different angle:

- Flow decomposition fails because the interaction appears at mover-equals-defect steps.
- Ambient topology fails because it does not see the interaction at all.
- Map-level fails because the interaction is pre-dynamical (no self-map exists).
- Sheaves fail because the interaction is global, not local-cellular.
- Arithmetic fails because the only separator is the threshold itself.

The $(C, \mu) \times \mathrm{detOf}$ interaction is named but not captured by any standard invariant class tested.

7.8 Currency reframing test

A separate question is whether the failure cascade of §§ 7.2 to 7.6 reflects a wrong choice of *currency*: we ask for a lower bound on $\prod m_i$, but $\prod m_i$ counts the ambient configuration space and does not directly measure what the dynamics constrains. A natural alternative is to measure on the cycle C itself: the mover-pattern period T_μ, the count κ of detOf-determined entries pinned by C, or the empirical entropy H of the cycle's value distribution. If any such quantity Q(system) separates the corpus more cleanly than $\prod m_i$ does, then the right form of the lower bound is $Q \geq q(n)$ and the product bound follows by counting.

We tested this on the 97-record stratified corpus of § 6.3 (10 valid witnesses + 87 sub-threshold candidates). For each record we computed T_μ, κ_{mover}, κ_{total}, H_{pos}, H_{joint}, and their natural normalizations. The summary (table 7) is unambiguous.

Caveat on the saturation column. The "saturation CV" comparison on $\log_2 \prod m_i$ is tautological: the 10 valid records were selected because they realize the conjectured saturating formula $M_n = 4 \cdot 3^{n-2}$, so of course they lie on a $\log_2 \prod m/n \approx 1.41$ curve. The 0.030 CV there is a definition, not a discovery, and the CV comparison is included only to show that no dynamical currency achieves tautological saturation on the valid side either. The substantive content of the table is in the AUC and strict-n columns, and in the direction flip discussed below.

Table 7: Per-currency separation of the 97-record corpus. AUC is the fraction of (valid, sub) pairs ordered consistently with the chosen direction. "Strict n" is the number of values of n (out of 5, since $n = 10$ has no sub-threshold record) at which the currency strictly separates valid from sub. Saturation CV is the coefficient of variation of $\overline{Q}/n^k$ across n, minimized over $k \in \{0, 1, 2\}$, on the valid side; smaller is better. Full per-n tables and methodology in the supplementary memo listed in appendix F.

currency Q	AUC	strict n	saturation CV
$\log_2 \prod m_i$	1.000	5	0.030
$\prod m_i$	1.000	5	1.048
$H_{\text{pos sum}}$	0.882	3	0.026
H_{joint}	0.866	2	0.029
T_μ	0.784	2	0.228
κ_{total}	0.782	2	0.114
κ_{mover}	0.697	2	0.128

The real content: direction flip at $n = 9$. Independently of the saturation-CV column, each dynamical currency *reverses direction* at $n = 9$. The valid record (CLB strip on $(2, 3, 3, 3, 3, 3, 3, 3, 2)$) has mover-pattern period $T_\mu = 3n - 2 = 25$, $\kappa_{\text{mover}} = 25$, $H_{\text{pos sum}} = 12.08$ (with good-configuration count $L = n^2 - 2n + 8 = 71$). The nine sub-threshold Table 7 records have $T_\mu \in \{39, \ldots, 53\}$, $\kappa_{\text{mover}} \in \{39, \ldots, 47\}$, $H_{\text{pos sum}}$ up to 12.53. Each dynamical currency we report, T_μ, κ, and H, *reverses direction*: on these measures the sub-threshold cycles are strictly longer, more mover-active, and more entropic than the valid one. Any attempt to lower-bound some monotone function of mover-pattern period, detOf-coverage, or entropy would flip the inequality at $n = 9$. Only $\prod m$ sees that these records are sub-threshold.

Observation 20 (Currency null)**.** On the 97-record stratified corpus, no candidate dynamical currency $Q \in \{T_\mu, \kappa, H, \text{their ratios}\}$ reorders valid vs. sub-threshold more cleanly than $\prod m_i$ does (same or worse AUC, same or worse strict-n count). More substantively, at $n = 9$ every candidate dynamical currency *reverses direction*; this is the statement with content (the AUC and saturation-CV columns are ancillary, the latter partly tautological by the saturating-family selection of the valid side). The direction flip rules out the reformulation "lower-bound a different, dynamical quantity": no monotone function of T_μ, κ, or H can lower-bound the sub-threshold side without flipping sign at $n = 9$. $\prod m_i$ is therefore the right ambient coordinate for the obstruction; the obstruction itself remains located in the joint $(C, \mu) \times \text{detOf}$ interaction (Observation 19).

This rules out the obvious reformulation, "solve the lower-bound problem on a different quantity." It does not rule out routes that extract a structural invariant of $X(\mathbf{m}) \setminus C$ while keeping $\prod m_i$ as the ambient coordinate; such routes are still open and indeed match the topological-invariant proof shape we find most natural for the problem.

8 Open questions

8.1 Analytical proof of T+sided-only circulation existence

Conjecture 21 (T+sided directed cycle)**.** *For every good cycle at $n \geq 9$ with product below $4 \cdot 3^{n-2}$, the subgraph of the lifted forced-NG graph induced by transport $\cup$ c_left $\cup$ c_right contains a directed cycle.*

A small-n piecewise variant can be stated for $n \in \{5, \ldots, 8\}$ at product below $32 \cdot 3^{n-4}$. If Conjecture 21 is proved together with the missing implication from a lifted directed cycle to peelTube-nonemptiness, the existing `SinkKernel.not_converges_of_SK_nonempty` machinery yields the conjectured large-n lower bound. The two implications are logically independent; both are currently open. The specific structural obstruction is the q-tube hole at mover-equals-defect steps (§ 7.2).

8.2 Uniform Axis C: sink-kernel nonemptiness on every sub-threshold candidate

Conjecture 22 (Uniform Axis C)**.** *For every candidate good cycle + partial-*detOf *record* $(\mathbf{m}, \text{cycle}, \text{movers}, \text{detOf})$ *produced by the enumeration pipeline at* $n \geq 9$ *with* $\prod m_i < 4 \cdot 3^{n-2}$*, the sink kernel* SK(detOf) *of the forced-NG digraph (§ 6.5) is nonempty.*

By § 6.5, a proof of Conjecture 22 would upgrade the one-sided certificate from the corpus-level empirical statement to a uniform-in-n structural theorem, discharging the large-n lower bound without the two-step (T+sided-only $\Rightarrow$ peelTube) routing. Axis C is not restricted to the Hamming-1 tube, and the SK operator is a discrete topological invariant of $X(\mathbf{m}) \setminus \text{cycle}$ (a closure property of the forced-move digraph), so the obligation is shaped as "a topological invariant forbids it" rather than an arithmetic statement on detOf. Supporting evidence at the first n beyond the exhaustive-certificate reach: the extended $n = 10, 11$ sweeps of § 6.5 fire Axis C on every one of the 291 sorted $n = 10$ sub-threshold multisets (320 probes, zero silent) and on 563 of 564 sorted $n = 11$ sub-threshold multisets (585 probes, zero silent, one SMT-budget UNKNOWN) across a two-tool (DFS + SMT) pipeline, extending the canonical-corpus verdict two levels past the exhaustive certificate at $n = 9$. Concrete openings flagged by the corpus audit: the four $n = 9$ non-adjacent $\{2^3, 3^5, 4\}$ records have nontrivial-SCC counts in $\{12, 16, 38, 12\}$ with largest SCCs of size ~ 1000, small enough for a direct analysis of their forced-orbit structure.

8.3 A pre-dynamical analog of Conley index

Does a map-level invariant exist that is well-defined when no valid self-map f exists? Candidate frameworks: partial-function Conley index; relation-valued homotopy. Such an invariant would discriminate self-stabilizability without requiring the system to be defined.

8.4 A sheaf formulation encoding global constraints

Construct a sheaf whose restriction maps encode convergence-propagation or single-priv-violation-propagation non-locally, and compute its H^1 on the 97-record canonical corpus. The standard local-cellular design space failed empirically (§ 7.5); a non-local sheaf could in principle access the global obstruction.

8.5 Extension of empirical verification

Extend the CLB or CUP-2 witness verification beyond $n = 18$, ideally by replacing finite verification with an all-n symbolic proof that both rule tables satisfy the six validity properties for every $n \geq n_0$. This would promote Conjecture 9 from a Python-verified finite range to an all-n theorem.

8.6 Lean formalization of the lower bound

Prove $M_n \geq 4 \cdot 3^{n-2}$ in Lean 4 / Mathlib for some $n \geq 10$, or for a restricted subclass of good cycles (e.g. those with longest_ter_run ≤ 1). This would move part of the lower-bound program from computational evidence toward proof-assistant-certified mathematics.

8.7 Relationship to ARG's 1985 binary-count conjecture

In what restricted settings does Conjecture 10 imply ARG's binary-count conjecture (no valid system has more than a constant number of binary processors as $n \to \infty$)? At ternary-dense $\{2, 3\}$ multisets the implication is immediate: Conjecture 10 forces $2^k \cdot 3^{n-k} \geq 4 \cdot 3^{n-2}$, hence $k \leq 2$. With higher-state compensation allowed the implication is not immediate.

8.8 Transport to the relaxed-connectedness convention

All bounds of § 3 are stated in the Dijkstra-connected model of § 2.2. The 1985 Stanford seminar worked under a relaxation that admitted multi-good-cycle constructions (including the $n = 4$ Gray-code family at product 16); write M_n^{rel} for the minimum state product under that convention. The first-data result $M_5^{\text{rel}} = M_5 = 96$ was stated in § 3.4 (theorem 12); this subsection records the underlying SMT table and the open questions beyond $n = 5$.

1. Is $M_n^{\text{rel}} = M_n$ for every $n \geq 5$?
2. For $n \in \{5, \ldots, 9\}$, is M_n^{rel} equal to the value in theorem 5, or strictly smaller (as it is at $n = 4$, where $M_4^{\text{rel}} \leq 16 < 24 = M_4$)?
3. Does the conjecture $M_n = 4 \cdot 3^{n-2}$ of Conjectures 9 and 10 transport to M_n^{rel}, and hence resolve Knuth's original 1985 questions as stated?

A positive answer to (1) would make the connected-model bounds of this paper directly applicable to Knuth's questions as originally stated; until then, the resolution offered by corollary 11 is of the connected-variant questions only. Theorem 12 answers question (1) positively at $n = 5$ and question (2) negatively at $n = 5$.

First data (randomized search). A standalone relaxed verifier (shipped with the repository bundle) confirms that every stored connected-valid witness ($w_{4\text{opt}}, w_5, w_6, w_7, w_8$) is also relaxed-valid (relaxed $\supseteq$ connected, as expected). Brute-force rule enumeration at $n = 5$ is infeasible (approximately 10^{10}–10^{12} rule systems per sub-threshold multiset), but a randomized sub-threshold sample of 5.5×10^6 rule systems distributed over five sub-target multisets ($\{2^5\}, \{2^4, 3\}, \{2^3, 3^2\}$ at $n = 5$; $\{2^6\}, \{2^5, 3\}$ at $n = 6$) yielded zero relaxed-valid hits. This is not a proof, but it is first negative data against $M_n^{\text{rel}} < M_n$ at $n \in \{5, 6\}$.

SMT table for $M_5^{\text{rel}} = M_5$. A z3 SMT encoding of the five relaxed-convention validity properties (with a `-sym-break` flag enabling WLOG quotients under the per-processor value-swap action and cyclic rotation of processor indices on (good, f)) certifies that *no relaxed-valid rule table exists at any of the five $n = 5$ sub-threshold multisets*:

multiset	$\prod m$	M_5	verdict	z3 time
$\{2^5\}$	32	96	UNSAT	0.02 s
$\{2^4, 3\}$	48	96	UNSAT	0.14 s
$\{2^4, 4\}$	64	96	UNSAT	8.94 s
$\{2^3, 3^2\}$	72	96	UNSAT	31.6 s
$\{2^4, 5\}$	80	96	UNSAT	90.2 s

Total sub-threshold runtime ≈ 130 s. An at-threshold sanity check at $\{2^3, 3, 4\}$ (product $96 = M_5$) returns SAT in 0.48 s and the extracted rule table passes the relaxed verifier end-to-end, confirming the encoding is correct. Combined, these certificates give $M_5^{\text{rel}} \geq 96$, and since $M_5^{\text{rel}} \leq M_5 = 96$ is automatic, $M_5^{\text{rel}} = M_5 = 96$ (theorem 12).

$n = 6$ **partial sweep.** Of the twelve $n = 6$ sub-threshold multisets, six are UNSAT-certified under the same encoding (every multiset with product ≤ 216, including the hardest-so-far $\{2^3, 3^3\}$ at 1905 s); the remaining six, at products $\{192, 192, 224, 240, 256, 256\}$, are in a batch-running sweep. The encoding source, per-multiset UNSAT logs, and findings memo are shipped with the repository bundle (appendix C.11).

9 Related work

9.1 Classical

Dijkstra [4] introduced self-stabilization; Tchuente [12] showed that in the uniform case ($m_i = m$ for all i) the common state count must satisfy $m \geq 3$, motivating Knuth's product-minimization generalization; Haddad–Knuth [8] (STAN-CS-85-1055) includes the seminar's $M_n > 2^n$ bound for $n > 4$, ARG's non-adjacent-binary LCM bound $L \geq N + 1$, and the quasi-unidirectionality bound; Dijkstra [5] gave the belated proof of Solution 3 correctness.

ARG's LCM bound remains the sharpest 1985-era structural constraint on the non-adjacent-binary sub-family and underpins both the all-binary impossibility reproof and the $k \leq 4$ binary cap in $\{2^k, 3^{n-k}\}$. As discussed in § 5.3, the bound does not extend to the adjacent-binary witnesses that realize M_n for $n \in \{4, \ldots, 8\}$.

9.2 Post-1985

Burns–Pachl [3] addressed uniform self-stabilizing rings without a distinguished processor; Ghosh [6] established the uniform 3-state lower bound under the central daemon with deterministic execution; Gouda–Haddix [7] gave the first stabilizing unidirectional deterministic token ring with a constant (8-state) per-process state count; Beauquier–Debas [1] gave an optimal 3-state algorithm for bidirectional non-uniform rings with $O(n^2)$ stabilization time; Hoepman [9] tightened bounds for Solution 1; Blin–Le Bouder–Petit [2] established the $O(\log \log n)$-bit memory threshold for self-stabilizing token circulation in a different model; Protocon [11] performs automated synthesis for the uniform-state version of the problem. None of these results addresses the non-uniform product-minimization question of § 2.5.

9.3 Lean and formal verification in distributed algorithms

Prior formal verification of self-stabilization includes a PVS treatment of Dijkstra's token ring, Coq frameworks for certified self-stabilization, and PADEC/Coq certifications of stabilization-time complexity. Relative to that prior work, the Lean 4 / Mathlib formalization underlying theorems 13 to 15 appears to be the first Lean 4 formalization of non-uniform state-product witnesses for Dijkstra's token-ring product-minimization problem at the scale considered here (state-count product up to 26244, ring size up to 10, with exhaustive per-n rank-table convergence certificates). The stated Lean theorems are sorry-free; finite per-n certificate tables are discharged by `native_decide` on sealed data.

10 Conclusion

We have determined M_n exactly for $n \in \{3, \ldots, 9\}$ (theorem 5): the rows $n \in \{3, 4\}$ are unconditional; the rows $n \in \{5, \ldots, 9\}$ are backed by the replayable rejection-certificate stream shipped under `artifacts/rejections/` in the release tag (appendix C.11). We have certified the sharp ratio crossover at $n = 9$ (remark 8) and stated the conjecture $M_n = 4 \cdot 3^{n-2}$ for all $n \geq 9$. The conjecture is supported by Lean-checked upper-bound constructions for $n \in \{4, \ldots, 10\}$, Python-verified constructions through $n = 18$, an exhaustive lower-bound certificate at $n = 9$, and structural detectors that separate the stratified corpus at $n \leq 9$. The circulation-LP detector separates every tested $n \le 8$ record (78 sub-threshold candidates LP-feasible and 8 $n \le 8$ valid witnesses LP-infeasible); at $n = 9$ it fires on 5 of 9 tested sub-threshold orderings of the three Table 7 multisets and returns LP-infeasible on the remaining 4, which structurally localize in the non-adjacent $k_{\text{bin}} = 3$ sub-family of $\{2^3, 3^5, 4\}$, the same sub-family on which ARG's 1985 LCM bound is also silent. A strict upgrade of the detector, Axis C (§ 6.5), takes the sink kernel of the candidate's forced-NG digraph and fires on all 87 sub-threshold records of the canonical 97-record corpus (including the four C1 blind-spot records) while remaining silent on every valid-witness record, delivering a one-sided sub-threshold certificate via monotonicity in detOf. Extended two-tool (DFS + SMT) sub-threshold sweeps at $n = 10, 11$ fire Axis C on every probe surfaced by the pipeline: 320 probes across all 291 sorted $n = 10$ sub-threshold multisets and 585 probes across 563 of 564 sorted $n = 11$ sub-threshold multisets (one SMT-budget UNKNOWN), zero silent. This is a per-probe statement, not a per-multiset completeness claim, see the scope caveat of § 6.5. Additionally, an SMT certificate shows $M_5^{\text{rel}} = M_5 = 96$ (theorem 12), the first case at $n \ge 5$ in which $M_n^{\text{rel}} = M_n$. See §§ 3.4, 6.4 and 6.5. We have located the analytical obstruction at the joint interaction of the good cycle, its mover sequence, and the determined rule-table entries, and have catalogued five families of failed analytical attempts on this interaction.

What we do *not* claim: a proof of the uniform lower bound or a proof of the all-n upper bound. The mechanism of the $n = 9$ crossover and the transport of our bounds to the relaxed-connectedness convention beyond $n = 5$ are discussed in §§ 3.4, 5.3 and 8.8. The state of the problem after this paper is: the small-n regime is exact, the crossover is certified, the large-n shape is conjectured with concrete supporting evidence, and the central open problem (Conjecture 21) is named with a specific structural obstruction.

Acknowledgments

The author thanks D. E. Knuth for the 1985 problem formulation and for correspondence that encouraged this write-up. The Lean 4 / Mathlib verification and the z3 SMT certificates reproduce on consumer hardware; the exhaustive lower-bound search at $n = 9$ was run on No Way Labs compute. The search emits a replayable rejection-certificate stream, archived under `artifacts/rejections/` in the release tag (appendix C.11); this is what a referee replays rather than re-running the full sweep. The rerun cost from a clean clone is hours on consumer hardware for the C1+C2 coverage manifest and a replay of any chosen certificate; a full re-enumeration is hours-to-days and is not required to verify any claim in this paper.

A Lean formalization details

The Lean 4 / Mathlib formalization lives in `lean/LeanMn/`. Build instructions:

```
cd lean
lake exe cache get
lake build LeanMn.UpperBound
lake build LeanMn.Main
```

The upper-bound-reachable build is sorry-free; finite per-n rank-table checks are discharged by `native_decide` on sealed certificates.

Main theorems referenced in the manuscript:

- `upper_bound_small`: in `LeanMn/UpperBound/Theorem.lean`: $M_n \le 32 \cdot 3^{n-4}$ for $n \in \{5, 6, 7, 8\}$.
- `upper_bound_cert`: in `LeanMn/UpperBound/Theorem.lean`: $M_n \le 4 \cdot 3^{n-2}$ for $n \in \{4, \dots, 10\}$ via the CUP-2 rule table.

- `M_4_eq_24`: in `LeanMn/SmallN/Theorem.lean`: $M_4 = 24$ exactly.

Auxiliary objects (`LeanMn/LowerBound/SK/`): the sink-kernel SK(gc) and the soundness theorem `SinkKernel.not_converges_of_SK_nonempty`; the Hamming-1 tube T_{N_1} and its peel peelTube(gc); the bridge theorems `sk_nonempty_via_tube_small_n` and `sk_nonempty_via_tube_large_n` which take peelTube(gc).Nonempty to SK(gc).Nonempty; the slab-counting infrastructure on SK(gc).

The CUP-2 rule table (87 entries, n-independent) appears in `LeanMn/Tables.lean`; the good cycle in `LeanMn/Cycle.lean`; the per-n rank certificates `cup2Converges4` through `cup2Converges10` in `LeanMn/SmallN/Cup2Convergence.lean`, auto-generated by `gen_cup2_ranks.py` (appendix F). The stated theorems are sorry-free; finite certificates are checked by `native_decide` on sealed tables.

A.1 Encoding audit: validity predicate versus prose definition

Because the stated Lean theorems discharge their finite content through `native_decide` on sealed certificates generated by auxiliary Python (`gen_cup2_ranks.py` and siblings), the correctness of the final Lean statement is only as strong as the correctness of the Lean-level *validity predicate* `IsValid` that encodes the six properties of § 2.3. A bug in `IsValid` (e.g. a missing clause, a swapped index, an off-by-one in the move relation) would allow a correctly-computed decision procedure to certify a mathematically-false theorem. The audit against this failure mode has three layers:

1. **Clause-by-clause transcription audit.** The Lean `IsValid` predicate is structured as a conjunction of six named sub-predicates, one per property of § 2.3, and each sub-predicate carries a docstring that quotes the prose definition of that property verbatim. An auditor can check clause-by-clause that each of (1)–(6) is realized in Lean as the conjunction-named sub-predicate claims to realize.

2. **Dual-path re-verification.** Every stored witness is independently re-checked by the Python verifier (`verify_witnesses.py`, `verifier.py`; appendix F), whose `verify_system` routine is a pure transcription of the six prose properties with no shared data structures with the Lean side. A disagreement between the two paths on any witness (Lean certifies valid, Python rejects; or vice-versa) is a run-time failure; no such disagreement has been observed on any stored witness across $n \in \{4, \ldots, 10\}$ or on the $n = 4$ non-witness $(2, 2, 2, 2)$ (which Python rejects, Lean rejects via `M_4_eq_24`'s lower-bound arm).

3. **Negative-control checks.** The same auto-generated rank-table machinery was run against deliberately-broken candidate witnesses (a rule-table entry with the wrong output, a cycle with a duplicated configuration, a cycle that skips a required firing) as sanity checks during development; each negative control failed the Lean `native_decide` call at the expected rank-table row. These logs are archived alongside the certificate stream in appendix C.11.

The audit trail is shipped as an automated cross-verification script (`parity_runner.py`) with reviewer-facing documentation indexed in appendix F. The runner executes all three audit layers as exit-code-returning steps:

- *Layer 1 (transcription).* Parses the Lean `IsValid` constituent rule tables (`wNM`, `wNP0..wN_Pk` in `LeanMn/SmallN/Defs.lean`; the five universal tables `TBotVal, TLowVal, TMidVal, THighVal, TTopVal` in `LeanMn/Tables.lean`) and compares them entry-by-entry against the Python encodings (`verify_witnesses.py::witness_nN`, for $N \in \{4, 4\text{opt}, 5, 6, 7, 8\}$; the five universal tables of `cup2_theorem.py`). Expected: zero diffs, with a SHA-256-truncated *rule-table hash* printed per witness that matches across encodings.

- *Layer 2 (dual-path re-verification).* Independently runs the Python `verify_system` on both the Lean-parsed table and the Python-loaded table; for the CUP-2 family also runs `verify_system` on `cup2_theorem.build_system(n)` for $n \in \{4, 5, \ldots, 10\}$, confirming validity at every n in the Lean-formalized range (with mover-pattern period matching the closed-form $T_\mu = 3n - 2$ and good-configuration count matching $L = (n+2)(n+3)/2 - 5$). The `--lake` flag additionally invokes `lake build LeanMn.SmallN.Defs`, whose success certifies that every `wN_valid` theorem still type-checks against the current Lean+Mathlib stack.

- *Layer 3 (negative controls).* For each stored witness, the runner mutates a single move-entry in a copy of the Lean-parsed rule table and re-invokes `verify_system`; the expected outcome is `valid=False` on every mutated copy.

A referee can rerun the cross-verification with a single command (`python3 parity_runner.py --lake`); the runner exits with a non-zero status equal to the number of disagreements, and prints per-step `PASS`/`FAIL` plus a machine-readable `parity_report.json` suitable for archival alongside the certificate stream of appendix C.11.

Representative output. A sample run against the current tree produces, abbreviated:

```
[L1 transcription] witness_n4    : 0 diffs   (hash 7c5d...)  PASS
[L1 transcription] witness_n4opt : 0 diffs   (hash a8a2...)  PASS
[L1 transcription] witness_n5    : 0 diffs   (hash 6651...)  PASS
[L1 transcription] witness_n6    : 0 diffs   (hash 163d...)  PASS
[L1 transcription] witness_n7    : 0 diffs   (hash a8a2...)  PASS
[L1 transcription] witness_n8    : 0 diffs   (hash 6e4b...)  PASS
[L1 transcription] CUP-2 universal tables : 0 diffs          PASS
[L2 dual-path]     verify_system across witnesses : 0/0 disagreements
[L2 dual-path]     build_system(n) for n=4..10     : valid=True (T_mu=3n-2)
[L2 dual-path]     lake build LeanMn.SmallN.Defs   : success
[L3 negative]      12/12 single-move-entry mutants : valid=False
----
parity_runner: 0 disagreements, exit status 0
```

Each `PASS` line corresponds to the hash column carried in the rule-table artifact of appendix E; a disagreement at any layer would be a run-time failure with a non-zero exit status.

B The circulation detector: formal construction

B.1 Lifted-graph vertex set

For a candidate good cycle $gc = [c_0, \ldots, c_{L-1}]$ on $\mathbf{m}$,

$$V_{\text{lift}}(gc) = \{(k, q, a) : k \in \text{Fin}(L),\ q \in \text{Fin}(n),\ a \in \text{Fin}(m_q),\ a \neq c_k(q),\ c_k[q := a] \in T_{N_1}(gc)\}$$

with projection $\pi(k, q, a) = c_k[q := a] \in T_{N_1}(gc)$.

B.2 Edge taxonomy

For each step $k \in \text{Fin}(L)$, let mov_k denote the firing processor. An edge $(k, q, a) \to (k', q', a')$ exists iff $k' = k + 1 \pmod L$ and $\pi(k', q', a') =$ apply-move of $\pi(k, q, a)$ at mov_k (under either the cycle's rule or a detOf-determined rule). The edge type is

- *transport* if $|\text{mov}_k - q| \geq 2 \pmod n$;
- *c_self* if $\text{mov}_k = q$;

- *c_left* if $\mathrm{mov}_k = q - 1 \pmod{n}$;
- *c_right* if $\mathrm{mov}_k = q + 1 \pmod{n}$.

The four types partition the forward edges. The audit of 4795 forward edges across the corpus found zero edges in any *other* bucket; the partition is complete in the forward direction.

B.3 LP formulation

$$\text{find } \Phi : E_{\text{lift}} \to \mathbb{Q}_{\geq 0} \text{ such that } B^\top \Phi = 0 \text{ and } \textstyle\sum_e \Phi_e = 1,$$

where B is the (vertex $\times$ edge) incidence matrix of the lifted-defect digraph. The positive-mass normalization $\sum_e \Phi_e = 1$ implements the $\Phi \neq 0$ condition as a linear constraint suitable for a standard LP solver.

Exactness. The LP is rational-valued by construction (B has $\{-1, 0, +1\}$ entries). The reported implementation uses a floating-point simplex for feasibility and then rationalises the returned solution; feasibility verdicts are post-checked by an exact incidence-balance pass (`exact_check.py`; appendix F) that refuses to accept a feasible verdict unless the rationalised Φ satisfies $B^\top \Phi = 0$, $\sum_e \Phi_e = 1$, and nonnegativity $\Phi_e \geq 0$ on every edge (zero-flow edges are allowed and common).

Infeasibility verdicts on the reported corpus are shipped as *integer-valued topological-sort certificates*, which are the Farkas witnesses for this LP form: for each of the 14 LP-infeasible records (four small-n absorbers w5..w8, six CLB ternary-strip $n \in \{5, \ldots, 10\}$, and the four $n = 9$ counterexamples of § 6.4), the artifact `farkas_certificates.json` (appendix F) contains an explicit vector $y : V_{\text{lift}} \to \mathbb{Z}_{\geq 0}$ (`topo_order`) such that for every edge $(s, t) \in E_{\text{lift}}$, $y[t] > y[s]$. Any non-zero non-negative circulation Φ would satisfy

$$0 = y^\top (B\Phi) = \sum_{e \in E_{\text{lift}}} \Phi_e \big(y[t(e)] - y[s(e)]\big) \geq \sum_e \Phi_e \cdot 1 > 0,$$

a contradiction; equivalently, the lifted-defect digraph is a DAG (topological sort exists iff no directed cycle). A standalone reviewer-runnable verifier (`farkas_verify.py`; appendix F) takes the certificate JSON, rechecks $y[t] > y[s]$ on every edge of every record, and exits with a status equal to the number of violations (expected 0). The certificate is purely combinatorial, no LP solver, no floating-point, no external dependencies.

B.4 Hoffman correspondence

Proposition 23. *The LP of appendix B.3 is feasible if and only if the lifted-defect digraph contains a directed cycle.*

Proof. This is Hoffman's circulation theorem [10]: a finite digraph has a non-trivial non-negative circulation iff it has a directed cycle. □ □

B.5 The 29-record corpus

Per-record details (state vector, n, class, LP status, edge-type flow statistics) are produced by `probe_wave3_combined_2026-04-26.py` and `probe_wave4_combined_2026-05-03.py` (appendix F). Table 8 gives the fully populated corpus (state vectors, products, lifted-graph edge counts, LP and restricted-LP verdicts, per-record flow-type support fractions, and per-record certificate hashes); a short representative sample appears in table 9.

B.6 Edge-type support statistics and perturbation tests

Selected-optimum flow statistics across the corpus average, in the orientation-invariant 3-type collapse, transport 66.5%, c_sided 33.5%, c_self 0% (§ 6.7); the cycle-orientation-dependent 4-type breakdown is c_right 33.2% / c_left 0.3%, which reversing the cycle swaps. The 11-variant perturbation suite (edge-class boundary perturbations, cycle reversal, alternative tube constructions) flipped no record's verdict.

B.7 Empirical vs proved

The detector's correspondence between LP feasibility and validity status is established empirically on the 29-record corpus. No extension of the corpus-separation claim beyond the 29 records or beyond $n \leq 10$ is asserted, and no theorem of the form "LP infeasible $\Rightarrow$ valid system exists" is asserted.

C Exhaustive verification methodology

The $\geq$ direction of theorem 5 at $n \in \{5, \ldots, 9\}$ is discharged by this appendix: the pre-generated rejection-certificate stream under `artifacts/rejections/` in the release tag witnesses every sub-target product-class elimination (appendix C.11).

C.1 Enumeration scheme

For each $n \in \{3, \ldots, 9\}$ and each target product $T < M_n$ (strictly below the claimed minimum):

1. enumerate every multiset $\mathbf{m}$ on n positive integers ≥ 2 with $\prod m_i = T$;
2. for each multiset, enumerate cyclic orientations modulo cyclic rotation, reflection, and (where applicable) state renaming;
3. for each oriented state-count vector, enumerate candidate good cycles;
4. for each candidate good cycle, enumerate consistent partial rule tables and full completions, applying the pruning rules of appendix C.2;
5. for each surviving completion, run the verifier of § 5.1.

C.2 Pruning rules

The search uses two proof-critical pruning rules, forced-neighbor consistency and processor-orientation orbit reduction, plus state-label-renaming canonicalization inside each candidate good cycle. A third rule, *quasi-unidirectionality* (no 4+ cyclically consecutive binary processors), is used only as an *optimization*: candidates pruned on this ground are also emitted in a dedicated replay class and the final verifier of § 5.1 rejects each of them independently. The lower-bound certificate of theorem 5 therefore does not depend on trusting quasi-unidirectionality as a lemma; it depends only on the verifier and on the two proof-critical pruning rules. Soundness proofs for the proof-critical rules are in appendix C.3; lemma 24 (the stand-alone quasi-unidirectionality statement) is recorded there for completeness but is explicitly flagged as non-proof-critical.

C.3 Soundness proofs of the pruning rules

1. **Quasi-unidirectionality (non-proof-critical).** The following statement holds in the six-property connected model of § 2.2 and is used by the search as an optimization only (per appendix C.2); the lower-bound certificate of theorem 5 does not depend on it, because the

final verifier of § 5.1 independently rejects every candidate with four cyclically consecutive binary processors and emits a replay-class rejection certificate.

Lemma 24 (Quasi-unidirectionality, connected model; optimization only)**.** *Let* $(\mathbf{m}, f_0, \ldots, f_{n-1})$ *be valid in the sense of § 2.3 (six properties including connectedness). Then* $\mathbf{m}$ *does not contain four cyclically consecutive binary processors: there is no index* j *with* $m_j = m_{j+1} = m_{j+2} = m_{j+3} = 2$ *(indices mod* n*).*

Lemma 24 is used in this paper *only* as a search optimization: candidates with four cyclically consecutive binary processors are emitted in a dedicated replay-class and independently rejected by the final verifier of § 5.1, so the lower-bound certificate of theorem 5 does not depend on the lemma. We therefore do not give a proof sketch here; the lemma is *not* proof-critical for any claim in this paper.

2. **Forced-neighbor consistency (proof-critical).** Each detOf entry is uniquely determined by the candidate good cycle (§ 2.8); inconsistency falsifies the cycle.

3. **Processor-orientation orbit reduction (proof-critical).** The dihedral action D_n (cyclic rotation + reflection) on processor positions is a homomorphism of the six-property connected model: the image of a valid system is again valid (a routine check on the definitions); hence one representative per D_n-orbit suffices.

4. **State-label renaming (proof-critical).** Let $\sigma = (\sigma_0, \ldots, \sigma_{n-1})$ with $\sigma_i \in \mathrm{Sym}(\mathrm{Fin}(m_i))$ be a tuple of per-processor state-label permutations. σ acts on configurations and rule tables by
$$(\sigma \cdot c)(i) \;=\; \sigma_i(c(i)), \qquad (\sigma \cdot f_i)(a, b, c) \;=\; \sigma_i\big(f_i(\sigma_{i-1}^{-1}(a), \sigma_i^{-1}(b), \sigma_{i+1}^{-1}(c))\big),$$
with indices modulo n.

Lemma 25 (State-label renaming is validity-preserving)**.** *The map* $(\mathbf{m}, f) \mapsto (\mathbf{m}, \sigma \cdot f)$ *is a bijection on systems with the same state-count vector. It carries good configurations of* f *to good configurations of* $\sigma \cdot f$ *(under* $c \mapsto \sigma \cdot c$*), and preserves all six validity properties of § 2.3. In particular, if the designer declares the good configurations of* $\sigma \cdot f$ *to be* $\{\sigma \cdot c : c \text{ good under } f\}$*, then* $(\mathbf{m}, f)$ *is valid iff* $(\mathbf{m}, \sigma \cdot f)$ *is.*

Proof sketch. The action on rule tables is defined so that $\sigma \cdot (f_i(c_{i-1}, c_i, c_{i+1})) = (\sigma \cdot f_i)(\sigma \cdot c)_{i-1,i,i+1}$; hence the privilege relation, the unique-move relation, and the move-output relation all commute with the action on configurations. Each of the six properties is expressed entirely in those terms, so it is preserved. Invertibility of the action (hence bijectivity) is immediate from the pointwise invertibility of each σ_i. Full details follow the same template as item 3; the complete clause-by-clause check is in `docs/state_label_renaming_proof.md` in the release tag (appendix F). □

State-label renamings are applied during candidate good-cycle canonicalization and counted separately from processor orientations (see appendix C.4, item C2).

C.4 Certificate format

For a referee to confirm that the enumeration is complete, each run of the search driver produces, alongside the rejection certificates, a *coverage manifest* consisting of the following five records.

(C1) **State-count coverage.** The list of every sorted multiset $\mathbf{m}$ on n positive integers ≥ 2 with $\prod m_i < M_n$, together with the enumeration loop bounds and a re-enumeration hash that must match a re-derivation from the integer-product constraints alone.

(C2) **Processor-orientation coverage.** For each multiset, the list of every dihedral orbit representative (cyclic rotation + reflection, i.e. D_n-orbit) of an ordered assignment of the multiset's values to processor positions; the orbit sizes sum to the ordered-tuple count $n!/\prod(k_v!)$, where k_v is the multiplicity of value v. Local state-label renaming (the separate action permuting the labels inside each $\mathrm{Fin}(m_i)$) is handled later during candidate-good-cycle canonicalization (C3) and does *not* enter the processor-orientation counts of Tables 10 and 11.

(C3) **Candidate good-cycle coverage.** For each oriented state-count vector, the list of candidate good cycles enumerated, with the pruning-rule invocations recorded per candidate.

(C4) **Partial rule-table coverage.** For each candidate good cycle, the list of partial rule-table completions considered up to the first violation or full confirmation.

(C5) **Pruning-rule soundness.** A self-contained restatement of each pruning rule together with a proof obligation discharged in the paper's model (appendix C.3).

Each rejected candidate additionally emits a *rejection certificate* consisting of: the state-count vector; the candidate good cycle (as a list of configurations); the partial rule table considered; and the specific violated property (with the witnessing configuration or move sequence that exhibits the failure). A certificate is accepted by an external checker iff the checker can independently reproduce the failing move sequence and verify that the emitted violation corresponds to one of the six properties of § 2.3.

C.5 Independent verifier

A second code path re-implements the verifier of § 5.1 from the published configuration-space definition without sharing internal data structures. It accepts the emitted certificates as input and re-checks each violation; discrepancies between the two checkers are flagged at run time.

C.6 Per-multiset enumeration at the $n = 9$ phase boundary

Tables 10 and 11 list every multiset class that the search eliminates at the boundary product $7776 = 32 \cdot 3^5$ and in the strict interval $(7776, 8748)$ at $n = 9$. For each multiset class, the search examines every *dihedral orbit representative* under D_9 (cyclic rotation + reflection of processor positions) and emits a failure certificate per orbit representative; soundness of the orbit reduction is supplied by appendix C.3, item 3. State-label renaming is handled separately inside candidate good-cycle canonicalization and is *not* folded into the orbit counts below. The orientation counts were computed by direct orbit enumeration under D_9 (order 18).

C.7 Per-n coverage summary

Table 12 is the per-n summary of the coverage manifest of appendix C.4. The number of sorted multisets below target is computed deterministically by the driver; the dihedral orbit representative count is computed exactly by orbit enumeration under D_n. The per-n C1+C2 rehash column is reproducible today from `coverage_manifest.json` (overall manifest hash `f684ed53216a0d7f`). The enumeration-dependent columns (candidate-good-cycle count, partial-rule-table count, rejections by pruning rule, rejections by final verifier) are populated from the full driver run and frozen into the pre-generated rejection-stream bundle shipped under `artifacts/rejections/` in the release tag; cells displayed as "†" below resolve to the corresponding `coverage_manifest.json` fields in that bundle (runtime envelope: $n \leq 8$ in minutes, $n = 9$ in hours–day; cf. appendix C.8). table 12 is the summary the referee needs in order to confirm the enumeration is complete; the per-certificate backing data live in the artifact.

C.8 Wall-clock and hardware notes at $n \in \{5, \dots, 9\}$

Per-n measured candidate-count and elapsed-time figures are produced by the search driver and saved to the project repository; the figures grow geometrically in the candidate space size, which itself grows substantially faster than the target product. This is the empirical basis for the $n = 10$ infeasibility statement of § 5.4. Per-row wall-clock and hardware data are included in the shipped rejection-stream bundle (appendix C.11).

C.9 Failure cases and sanity checks

Sanity checks applied throughout: (i) every stored valid witness is re-verified by the same verifier; (ii) every rejected candidate is re-rejected by the independent verifier; (iii) the small-n regime boundary check ($n = 4$, target 24, witness $\mathbf{m} = (2, 2, 2, 3)$) is run jointly against the Lean `M_4_eq_24` formalization and the Python verifier and must agree.

C.10 Reproducibility

The driver bundle (`driver.py` / `multiset_enum.py` / `generate_manifest.py` of appendix F.2, emitting `coverage_manifest.json` of appendix F.3) is deterministic and has no randomness; the emitted certificate stream is reproducible bit-for-bit. The inner C3 + C4 + C5 loop is also available as a compiled C worker (`exhaustive.c` in the same directory; build with `make`) whose aggregate output is bit-identical to the Python driver's modulo the canonical starting configuration chosen for each cycle (Python's lex-minimum vs. the C worker's mixed-radix hash-minimum; both enumerate the same dihedral-orbit representatives of candidate good cycles).

The C1 + C2 portions of the coverage manifest (state-count coverage and processor-orientation coverage) populate from a clean clone in seconds and match the deterministic columns of table 12: 0, 1, 5, 12, 27, 59, 147 sub-target multisets and 0, 1, 6, 22, 77, 343, 2085 D_n-orbit representatives at $n = 3, \dots, 9$, with SHA-256-truncated manifest hash `f684ed53216a0d7f`.

The C3 + C4 + C5 populations and the rejection-certificate stream are produced by running the driver (or the C worker) end-to-end. Wall-clock and hardware footprints per n are reported in appendix C.8.

The ship-state of `artifacts/rejections/` in the release tag is:

- full per-certificate stream for $n \in \{3, 4\}$ (`n3/`, `n4/`);
- full per-certificate stream for the all-binary orientation at $n = 5$ (`n5/ms-32-2-2-2-2-2.jsonl`), as a worked example of the mechanism at the smallest orientation where ternary branching does not yet blow up completion counts;
- no pre-generated stream for the remaining five $n = 5$ dihedral orbit representatives (products 48, 64, 72, 72, 80) or for $n \in \{6, \dots, 9\}$; per-cert streams at those $(n, \text{orientation})$ pairs are regenerable with the C worker as specified below, with per-orientation wall-clock on a single laptop core ranging from minutes at $n = 5$ binary-dominated orientations to hours at mixed orientations at $n \geq 6$; per-orientation bundle sizes grow with the free-entry completion product $\prod_{\text{free}} m_p$ and can reach 100 GB–1 TB of JSONL at worst-case multisets, which is why pre-generation is not repo-shippable for every $(n, \text{orientation})$.

For any $(n, \text{orientation})$ whose pre-generated stream does not ship, a referee can regenerate the full stream by building the C worker and invoking it per orientation:

```
cd probes/exhaustive_search
make
./exhaustive --n N --ms <sorted ms> --orient <orient> > out.jsonl
python3 driver.py --replay out.jsonl
```

Regeneration is deterministic; any discrepancy in aggregate rejection count against what the corresponding `summary.json` / `nN/index.json` row asserts is a reportable finding.

C.11 Artifact archive

The driver scripts, the coverage manifest of appendix C.4, the pre-generated rejection-certificate stream for $n \le 9$, and the $n \le 10$ detector corpora are packaged as a persistent artifact (Zenodo DOI pinned at submission / GitHub release tag); all repository paths listed in appendix F are relative to that archived snapshot. The pre-archive working tree is the project repository cited throughout. The release is tagged `v1.0` at submission; the DOI is the Zenodo mint of that tag.

D Detailed failed-approach descriptions

This appendix expands § 7. For each tested approach we record the formal statement of the proposed invariant; the pre-registered discrimination criterion (the empirical signal whose presence would have promoted the route to a candidate proof); the verdict of the probe; the specific obstruction that triggered the verdict; and the data link. Following the structure of the project's lower-bound history document (listed in appendix F, §§1–10), the catalog covers the pre-SK case-split architecture, the SK pivot, the five-route SK arc, the transport-lift / twist-calculus arc, the peel-direct approach, the A1 target-injectivity approach, the SK Morse–Hamming formulation, and the sequence of topological probes surveyed in § 7. Each entry's contents follow the template of § 7.1.

D.1 Pre-SK case-split architecture

Architecture: case split on cycle type given ≥ 3 binary and sub-threshold product. Six cases (safe processor, ZW cw=0, ZW cw>0, sweep consecutive, sweep non-consecutive, odd winding) with cross-case dispatchers. Verdict: stalled at 2–4 sorries; mutual recursion in the proof logic prevents a closed Lean proof; the architecture is fundamentally case-split. The 3-consecutive-binary case at $n \ge 9$ is itself an open analytical problem (no known mechanism).

D.2 SK pivot and the 10-edge skeleton

Architecture: prove that the binary-cube projection of $\mathrm{SK}(C)$ at sub-threshold $n \ge 9$ contains a canonical 10-edge skeleton. Verdict: the 10-edge form failed empirically at maximally-spread binary placements; replaced by a 4-pole form; the audit retiring the quantitative $|\mathrm{SK}| \ge 2^{n-1}$ target made the skeleton machinery no longer needed for the main argument.

D.3 The five-route SK arc

Five structurally distinct routes were attempted on the $\mathrm{SK}(gc)$.Nonempty target: direct sync-cascade (reproduces the A1-injectivity wall at step level); orbit-level quotient (collapses to A1); fiber-counting / fiber-budget (routes through A1); read-2 aggregate ansatz (independent quantitative wall at $n \ge 8$ binary-dominated); and abstract cycle-decomposition attempts (jointly: the obstruction lives in the $(C, \mu) \times \mathrm{detOf}$ interaction, not in either component alone). The named conjecture surviving the arc is the strict-injectivity sibling of Conjecture 21; empirical support is substantial (no violator on ~ 16.9M terminals across 26 seeds in the three-consecutive-binary regime) but no analytical proof is known.

D.4 Transport-lift / twist-calculus arc

Transport-lift arc: the forest bound was refuted; a directed-cycle survives in the tube digraph D_{tube}; the tube-index cocycle is multi-valued; and a structural "minimum Case-C = 6" observation holds across $n \geq 6$. The twist-calculus package (Dominant / Fold / Fusion regimes) was designed to force a closed escape-free walk on D_{tube}; empirically the escape rate grows linearly from 0.36 to 0.43 as n increases, eliminating the forced-closure route.

D.5 Peel-direct approach

Empirical floor excellent (100% peel-nonempty across 1898 records; exactly one non-trivial SCC per record; minimum SCC size grows with n). Analytical wall: peel existence is a global property; no uniform deterministic walk rule simulates it (26% residue on best-of-five rules tested); no local predicate characterizes SCC membership (min accuracy 46.7%); Mathlib has no partial-function cycle-existence lemma at this shape. Realistic a direct bounded case-split formalization appears possible but would likely be large and brittle.

D.6 A1 target-injectivity approach

Empirical floor: 6289 data points, zero violators. Caller audit broken: under unique-privilege assembly, mover and non-mover at the same context cannot be assembled. Moore pigeonhole fails: orphans live outside the cycle. HKR Index Lemma: structural preconditions absent (oriented manifold with boundary, rank-symmetry group action, chromatic subdivision). Literature search exhausts the off-the-shelf options.

D.7 SK Morse–Hamming arc

Conjecture: $f \supseteq \mathrm{detOf}(C)$ self-stabilizing iff f induces a complete discrete Morse vector field on the complement of C with critical cells in $G(C)$. Probe-level verdicts on ten invariants: Betti numbers either always zero, ambient-dimension-driven, or inverted via coverage scaling. Honest verdict: the LB has a combinatorial core that any topological invariant of the ambient or complement spaces is either (n, L)-driven, coverage-driven (wrong direction), or reducible to SK via sink iteration.

D.8 Topological probes

A systematic probe campaign covered, in order, six groups of topological-style invariants.

1. **Ambient invariants.** Four of four tested invariants (π_1, linking matrix, Lefschetz numbers, Cheeger constant) did not discriminate sub-threshold from valid-witness records.

2. **Lifted-defect circulation route.** A candidate separator. Empirically robust across the full corpus but analytically blocked at the mover-equals-defect step (§ 7.2); the separator was also used to produce a permanent positive side-effect, a Lean threshold bug was fixed by splitting the previously unprovable `peelTube_nonempty` into the small-n and large-n arms matching definition 2.

3. **Balance-identity hardening.** The direction-covariant 3-type balance identity closed to within a 4.9% leading-order residual, short of the 1% discrimination target.

4. **Decisive small-n tests and arithmetic pivot.** Both the 4-type and the direction-covariant 3-type balance identities failed as separators; the arithmetic-inequality pivot is decisive (best extracted feature is the threshold condition itself).

5. **Restricted-class candidate.** One restricted-class candidate survived as a conjecture on good cycles with longest_ter_run ≤ 1: if no two cyclically consecutive m_i entries both equal 3, the lifted circulation LP admits a feasible flow supported on transport + sided-context $(c_{\text{left}}, c_{\text{right}})$ edges alone, with no c_{self} edges used. The extended check

(`longest_ter_run_restricted_check.py`; appendix F) confirms this on all 58/58 in-subclass records of the $n = 5$–8 sub-threshold corpus plus the $n = 9$ Table 7 stretch (68 records total); the 4 LP-infeasible records at $n = 9$ are precisely the non-adjacent $\{2^3, 3^5, 4\}$ orientations, all of which lie outside the subclass (longest_ter_run $\in \{3, 4\}$). Two further probes were inconclusive.

6. **Nerve and non-standard sheaves.** A proper Čech H^1 computation via the nerve of the cover produced β_1 in the range $[27, 11179]$ at valid-witness records, not zero; tube refinements, transport-plus-sided-only circulation, Conley-index $\beta_1/|N|$ ranges, and three non-standard sheaf formulations all failed to separate the corpus.

Each group was retired on a pre-committed empirical or analytical kill criterion.

E Small-n witness data

State vectors, good cycles, and rule tables for the small-n absorber witnesses `witness_n5` through `witness_n8` and the CLB ternary-strip witnesses for $n \in \{5, \dots, 18\}$ live in the Python bundle indexed by appendix F.2: `verify_witnesses.py` (small-n absorber tables for $n \in \{4, \dots, 8\}$); `clb_witness_8748.py` ($n = 9$ CLB witness builder); `clb_inherent_cycles.py` (CLB sweep for $n \in \{5, \dots, 18\}$); and `gen_cup2_ranks.py` (per-n CUP-2 rank certificates for the Lean formalization).

The $n = 5$ small-n absorber witness is illustrative. Take $\mathbf{m} = (2, 2, 2, 3, 4)$ with P_0, P_1, P_2 binary, P_3 ternary, P_4 quaternary; the binary block sits at positions 0–2 and the quaternary at position 4, with the ternary buffer P_3 at position 3. The verified good cycle has length $L = 18$ and visits every processor at least once. The full transition tables and the cycle in configuration form appear in the witnesses memo listed in appendix F.

F Supplementary-files manifest

Every file referenced by name in the body is indexed here in one place, organized by type. All paths are relative to the archived snapshot of appendix C.11; in the body we use short file names and point here for the path.

F.1 Lean 4 / Mathlib source modules

- `lean/LeanMn/SmallN/Theorem.lean`: small-n theorems, including `M_4_upper`, `M_4_lower`, `M_4_eq_24` (theorem 15 and § 4.1).
- `lean/LeanMn/SmallN/Defs.lean`: per-witness rule tables `wNM`, `wN_P0..wN_Pk` for $N \in \{4, 4\text{opt}, 5, 6, 7, 8\}$ (appendix A.1).
- `lean/LeanMn/SmallN/Cup2Convergence.lean`: per-n CUP-2 rank certificates `cup2Converges4`–`cup2Converges10` (theorem 14).
- `lean/LeanMn/UpperBound/Theorem.lean`: `upper_bound_small` and `upper_bound_cert` (theorems 13 and 14).
- `lean/LeanMn/Tables.lean`: the five universal CUP-2 rule tables `TBotVal`, `TLowVal`, `TMidVal`, `THighVal`, `TTopVal` (§ 4.4).
- `lean/LeanMn/Cycle.lean`: the good cycle on the ternary-strip state vector (§ 4.4).
- `lean/LeanMn/LowerBound/SK/`: sink-kernel infrastructure, including `not_converges_of_SK_nonempty`, `sk_nonempty_via_tube_small_n`, `sk_nonempty_via_tube_large_n`, and the slab-counting apparatus (§ 6.5 and appendix A).

- `lean/LeanMn/LowerBound/`: umbrella directory for the lower-bound program (§ 2.8).

F.2 Python drivers, verifiers, and probes

- `probes/exhaustive_search/driver.py`: entry point for the exhaustive lower-bound search (appendix C.10).
- `probes/exhaustive_search/multiset_enum.py`: C1+C2 state-count and processor-orientation enumerator (appendix C.10).
- `probes/exhaustive_search/generate_manifest.py`: aggregator emitting `coverage_manifest.json` (appendix C.10).
- `docs/verify_witnesses.py` / `docs/verifier.py`: independent Python verifier whose `verify_system` routine re-checks every stored witness and every rejected candidate certificate (§ 5.1 and appendices A.1, C.5 and E).
- `probes/clb_inherent_cycles.py`: CLB ternary-strip sweep for $n \in \{5, \dots, 18\}$, covering up to 172M configurations at the largest tested n (§ 4.3 and appendix E).
- `probes/clb_witness_8748.py`: $n = 9$ CLB witness builder (appendix E).
- `probes/gen_cup2_ranks.py`: generator for per-n CUP-2 rank certificates consumed by `Cup2Convergence.lean` (appendices A and E).
- `probes/cup2_theorem.py`: Python mirror of the CUP-2 rule tables and `build_system(n)` (appendix A.1).
- `docs/parity/parity_runner.py`: three-layer Lean/Python cross-verification runner (appendix A.1).
- `docs/topo_revival/exact_check.py`: rationalising LP post-check for the circulation detector (appendix B.3).
- `docs/topo_revival/farkas_verify.py`: reviewer-runnable Farkas-certificate verifier (appendix B.3).
- `docs/arg_lcm/arg_lcm_finite_checks.py`: dihedral enumeration of the non-adjacent orientations of $\{2^3, 4, 3^{n-4}\}$ at $n = 8, 9$ and the four LCM-functional readings (§ 5.3).
- `probes/topo_revival/wave3/probe_wave3_combined_2026-04-26.py`: wave-3 probe driver producing the circulation-LP corpus (appendix B.5).
- `probes/topo_revival/wave4/probe_wave4_combined_2026-05-03.py`: wave-4 probe driver producing `phaseW4_results.json` (appendix B.5).
- `docs/axisc/probe_axisc_n10_sweep.py`: Axis C DFS enumerator for the $n = 10$ sub-threshold sweep (§ 6.5).
- `docs/axisc/probe_cycle_smt.py` / `docs/axisc/sweep_smt_n10_uncovered.py`: SMT cycle-existence prober and SMT sweep driver; reused at $n = 11$ by redirecting `-in` / `-out` to the $n = 11$ sweep inputs, closing the 214 budget-limited $n = 10$ multisets and 473 of the 474 budget-limited $n = 11$ multisets (§ 6.5).
- `docs/topo_revival/longest_ter_run_restricted_check.py`: restricted-class LP check on records with longest_ter_run ≤ 1 (appendix D.8).

- `docs/m5rel/smt_relaxed_n5_n6.py`: z3 SMT encoding of the relaxed-connectedness validity properties and the `-sym-break` WLOG quotients (theorem 12 and § 8.8).
- `docs/m5rel/relaxed_verifier.py`: standalone relaxed-convention verifier for stored connected-valid witnesses (§ 8.8).

F.3 Data, corpus, and certificate artifacts

- `corpus_canonical.json`: 97-record canonical corpus carrying LP and Axis C verdicts per row; overall SHA-256 hash `f4b017b1f57687cc` (table 6 and § 6.5).
- `corpus.json`: 29-record stratified sub-corpus referenced in the per-record hash recipe (table 8).
- `phaseW4_results.json`: wave-4 probe driver output backing the populated corpus table (table 8).
- `docs/topo_revival/farkas_certificates.json`: integer-valued topological-sort (Farkas) certificates for the 14 LP-infeasible records (appendix B.3).
- `parity_report.json`: machine-readable per-step output of `parity_runner.py` (appendix A.1).
- `coverage_manifest.json`: C1+C2 + C3+C4 coverage records of the exhaustive lower-bound search; overall SHA-256 hash `f684ed53216a0d7f` (table 12 and appendices C.4 and C.10).
- `docs/axisc/axc_n10_sweep_results.json`: Axis C DFS sweep results over 291 sub-threshold multisets at $n = 10$ (§ 6.5).
- `axc_n10_smt_sweep_results.json`: SMT-closed sweep over the 214 budget-limited $n = 10$ multisets (§ 6.5).
- `docs/axisc/axc_n11_sweep_results.json`: Axis C DFS sweep results over 564 sub-threshold multisets at $n = 11$ (§ 6.5).
- `axc_n11_smt_sweep_results.json`: SMT-closed sweep over the 474 budget-limited $n = 11$ multisets (473 FOUND, 1 UNKNOWN) (§ 6.5).
- `docs/axisc/axis_c_forced_ng_full_results.json`: earlier 72-record slice of the Axis C verdict preceding the canonical corpus (§ 6.5).
- `relaxed_search_n5_n6.json`: output of the 5.5×10^6 randomized relaxed-search sample at $n \in \{5, 6\}$ (§ 8.8).

F.4 Markdown memos and notes

- `docs/axisc/phase1_n8_summary.md`: $n = 8$ strengthening-effort summary behind the full-sweep sub-threshold corpus and the $n = 9$ Table 7 additions (§§ 6.3 and 6.4).
- `docs/axisc/n9_detector_design.md`: design-space catalog for an $n \geq 9$ detector on the non-adjacent $k = 3$ binary sub-family; restricted-to-edge-type construction marked RED, two alternative axes open (§ 6.4).
- `docs/axisc/n9_detector_findings.md`: consolidated eight-probe-axis findings underpinning the Axis C upgrade (§ 6.5).
- `docs/axisc/axisc_n10_n11_findings.md`: combined memo covering the $n = 10$ DFS and SMT sweeps under Axis C and their $n = 11$ extension.

- `docs/corpus_reconciliation.md`: reconciliation pass filling the 25 records outside the wave-1 Axis C slice (§ 6.5).
- `docs/m5rel/smt_relaxed_findings.md`: findings memo covering the M_5^{rel} SMT certificate and the $n = 6$ partial sweep (§§ 3.4 and 8.8).
- `docs/currency/phase1_verdict_2026-04-23.md`: per-n tables and methodology behind the currency-reframing verdict (table 7).
- `docs/lean_docs/lb_all_paths_history.md`: glossary (§0.1) of structural objects used in § 2.8, and the §§1–10 catalog of lower-bound paths attempted that drives appendix D.
- `docs/state_label_renaming_proof.md`: the complete clause-by-clause proof of lemma 25 (state-label renaming is validity-preserving).
- `witnesses.md`: full transition tables and good cycles in configuration form for the small-n absorber witnesses (appendix E).
- `parity_runner_README.md`: reviewer-facing documentation for the parity runner of appendix A.1.
- `docs/arg_lcm/paper.md`: self-contained dispatch of the ARG-LCM analysis used in § 5.3.

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Table 8: The 29-record stratified corpus, populated from `phaseW4_results.json` of the wave-4 probe driver (§ 6.3). Rows partition into 19 sub-threshold candidates (8 at $n = 5$, 7 at $n = 6$, 4 at $n = 7$), 4 small-n absorbers, and 6 ternary-strip (CLB) witnesses. Columns: record id; n; state vector $\mathbf{m}$; record class (sub = sub-threshold candidate, abs = small-n absorber, tri = ternary strip); $\prod_i m_i$; threshold B_n; source of the good cycle gc (enum = `enumerate_candidate_good_cycles`, witness = stored valid system); LP status; restricted LP status with c_self edges forbidden; lifted-graph edge count $|E|$; support fractions on the selected LP optimum in the order (transport, c_right, c_left, c_self), given as percentages; per-record certificate hash (8-char SHA-256 prefix over $(n, \mathbf{m}, \prod_i m_i, |E|, \text{feasible}, \text{supp\_edge\_type\_hist})$ in canonical JSON). Support fractions for sub-threshold records are taken on the default-solver optimum; c_self support being 0.00 corresponds to the secondary observation in § 6.7, and the restricted-feasibility status does not depend on which optimum is chosen. Edge-weighted pooled means across the 19 sub rows: transport 65.7%, c_right 33.2%, c_left 0.3%, c_self 0.8% (matching the headline figures of § 6.7); per-record means: 60.3/37.6/0.8/1.3. Overall 29-record LP-detailed sub-corpus hash of the canonical 97-record corpus (16-char SHA-256 prefix): `98faf429b06e5f92`; the full 97-record artifact is `corpus_canonical.json` (overall hash `f4b017b1f57687cc`; appendix F).

#	n	$\mathbf{m}$	cls	$\prod m$	B_n	gc	LP	LP$^{-\text{self}}$	$\|E\|$	flow %	hash
Sub-threshold candidates (19; all LP feasible)											
1	5	$(2,2,2,2,2)$	sub	32	96	enum	f	f	40	33.3/66.7/0.0/0.0	`409d60b2`
2	5	$(2,2,2,2,4)$	sub	64	96	enum	f	f	40	33.3/66.7/0.0/0.0	`cb2704f8`
3	5	$(2,2,2,3,2)$	sub	48	96	enum	f	f	60	50.0/50.0/0.0/0.0	`20d5eaef`
4	5	$(2,2,2,4,2)$	sub	64	96	enum	f	f	60	57.1/42.9/0.0/0.0	`648c3c2c`
5	5	$(2,2,3,2,2)$	sub	48	96	enum	f	f	60	42.9/50.0/7.1/0.0	`490e6659`
6	5	$(2,2,3,3,2)$	sub	72	96	enum	f	f	76	50.0/43.8/0.0/6.2	`14e72cb2`
7	5	$(2,2,5,2,2)$	sub	80	96	enum	f	f	83	61.1/33.3/0.0/5.6	`1f1356f2`
8	5	$(2,3,2,2,3)$	sub	72	96	enum	f	f	60	50.0/42.9/7.1/0.0	`52ef153f`
9	6	$(2,2,2,2,2,2)$	sub	64	288	enum	f	f	60	41.7/58.3/0.0/0.0	`12494377`
10	6	$(2,2,2,2,6,2)$	sub	192	288	enum	f	f	119	77.5/22.5/0.0/0.0	`106266e2`
11	6	$(2,2,2,5,2,2)$	sub	160	288	enum	f	f	124	67.5/32.5/0.0/0.0	`214b19b5`
12	6	$(2,2,3,3,3,2)$	sub	216	288	enum	f	f	145	68.2/27.3/0.0/4.5	`7c5cdd57`
13	6	$(2,3,3,2,3,2)$	sub	216	288	enum	f	f	168	59.1/31.8/0.0/9.1	`bf466e5b`
14	6	$(2,5,2,3,2,2)$	sub	240	288	enum	f	f	145	69.0/31.0/0.0/0.0	`deae2c28`
15	6	$(3,2,2,4,2,2)$	sub	192	288	enum	f	f	105	73.5/26.5/0.0/0.0	`526cc40d`
16	7	$(2,2,2,2,2,2,2)$	sub	128	864	enum	f	f	84	75.0/25.0/0.0/0.0	`58165611`
17	7	$(2,3,2,2,5,2,3)$	sub	720	864	enum	f	f	125	79.4/20.6/0.0/0.0	`6ef77873`
18	7	$(2,4,2,3,2,2,3)$	sub	576	864	enum	f	f	125	79.4/20.6/0.0/0.0	`4c8a499b`
19	7	$(3,2,2,2,2,3,4)$	sub	576	864	enum	f	f	110	78.1/21.9/0.0/0.0	`072a5a36`
Small-n absorber (4; all LP infeasible, sharp at their n)											
20	5	$(2,2,2,3,4)$	abs	96	96	witness	i	n/a	142	n/a	`6651df35`
21	6	$(2,2,2,4,3,3)$	abs	288	288	witness	i	n/a	282	n/a	`163da56f`
22	7	$(3,2,2,2,3,4,3)$	abs	864	864	witness	i	n/a	529	n/a	`a8a25d37`
23	8	$(2,2,3,4,3,3,2,3)$	abs	2592	2592	witness	i	n/a	748	n/a	`6e4bf101`
Ternary-strip (6; all LP infeasible, sharp only at $n = 9$)											
24	5	$(2,3,3,3,2)$	tri	108	96	witness	i	n/a	70	n/a	`1e3c1971`
25	6	$(2,3,3,3,3,2)$	tri	324	288	witness	i	n/a	113	n/a	`ce483a81`
26	7	$(2,3,3,3,3,3,2)$	tri	972	864	witness	i	n/a	168	n/a	`d2b44411`
27	8	$(2,3,3,3,3,3,3,2)$	tri	2916	2592	witness	i	n/a	235	n/a	`6fa61731`
28	9	$(2,3,3,3,3,3,3,3,2)$	tri	8748	8748	witness	i	n/a	314	n/a	`63477fa4`
29	10	$(2,3,3,3,3,3,3,3,3,2)$	tri	26244	26244	witness	i	n/a	405	n/a	`eed54a4a`

Notes. "f" = feasible, "i" = infeasible, "n/a" = the restricted-LP column only applies to LP-feasible records; "LP$^{-\text{self}}$" is the LP of appendix B.3 with c_self edges deleted from E_{lift}. The "sub" stratification at each n spans composition classes from near-all-binary up through the immediate sub-threshold anchor; the exact enumeration is deterministic from the probe-driver seed. Per-record hashes are reproducible from `corpus.json` via the canonical-JSON / SHA-256 recipe given in the caption.

Table 9: Representative corpus records (sample of the 29).

class	n	$\mathbf{m}$	LP
sub-threshold	5	$(2,2,2,2,3)$	feasible
sub-threshold	5	$(2,2,2,3,3)$	feasible
sub-threshold	6	$(2,2,2,3,2,3)$	feasible
sub-threshold	7	$(2,2,2,2,3,3,3)$	feasible
small-n absorber	5	$(2,2,2,3,4)$	infeasible
small-n absorber	6	$(2,2,2,4,3,3)$	infeasible
small-n absorber	7	$(3,2,2,2,3,4,3)$	infeasible
small-n absorber	8	$(2,2,3,4,3,3,2,3)$	infeasible
ternary-strip (CLB)	9	$(2,3,3,3,3,3,3,3,2)$	infeasible
ternary-strip (CUP-2)	10	$(2,3,3,3,3,3,3,3,3,2)$	infeasible

Table 10: All multisets at $n = 9$ with $\prod_i m_i = 7776$ and each $m_i \geq 2$; all three are eliminated by the search. Orientation count is modulo cyclic rotation and reflection (dihedral group D_9).

multiset	orientations
$\{2^3, 3^5, 4\}$	28
$\{2^4, 3^4, 6\}$	38
$\{2^5, 3^3, 9\}$	28
total orientations	94

Table 11: All 24 multisets at $n = 9$ with $\prod_i m_i \in (7776,\ 8748)$ and each $m_i \geq 2$; all are eliminated by the search.

multiset	$\prod_i m_i$
$\{2^8, 31\}$	7936
$\{2^8, 32\}$	8192
$\{2^8, 33\}$	8448
$\{2^8, 34\}$	8704
$\{2^7, 3, 21\}$	8064
$\{2^7, 3, 22\}$	8448
$\{2^7, 4, 16\}$	8192
$\{2^7, 4, 17\}$	8704
$\{2^7, 5, 13\}$	8320
$\{2^7, 6, 11\}$	8448
$\{2^7, 7, 9\}$	8064
$\{2^7, 8^2\}$	8192
$\{2^6, 3^2, 14\}$	8064
$\{2^6, 3^2, 15\}$	8640
$\{2^6, 3, 4, 11\}$	8448
$\{2^6, 3, 5, 9\}$	8640
$\{2^6, 3, 6, 7\}$	8064
$\{2^6, 4^2, 8\}$	8192
$\{2^6, 5^3\}$	8000
$\{2^5, 3^3, 10\}$	8640
$\{2^5, 3^2, 4, 7\}$	8064
$\{2^5, 3^2, 5, 6\}$	8640
$\{2^5, 4^4\}$	8192
$\{2^4, 3^3, 4, 5\}$	8640

Note. $\{2^5, 4^4\}$ denotes five copies of 2 and four copies of 4 (nine elements total), product $8192 = 2^5 \cdot 4^4$. All other rows use the analogous exponent notation. The complete per-orientation elimination certificates are emitted by the search driver (appendix C.10) and archived alongside the source (appendix C.11). The above is the exhaustive list of 24 multisets at $n = 9$ with product strictly between 7776 and 8748 and all $m_i \geq 2$; it is closed under the integer-product constraint, so no further class is missed.

Table 12: Per-n coverage summary of the exhaustive lower-bound search. The deterministic columns ("multisets $< M_n$", "D_n orbit reps") are emitted by `generate_manifest.py` and are reproducible from `coverage_manifest.json` (overall C1+C2 manifest hash `f684ed53216a0d7f`; per-n C1+C2 rehash in the last column). The four enumeration-dependent columns (candidate good cycles, partial rule tables, rejections by pruning, rejections by final verifier) are generated by running the driver end-to-end against the release tag (cf. appendix C.10; runtime envelope: $n \leq 8$ in minutes, $n = 9$ in hours–day). A pre-generated bundle of the full enumeration at $n \leq 9$ is shipped under `artifacts/rejections/` in the release tag; the values populated in the "†" cells below match that bundle row-for-row. The $n = 3$ row is closed today (zero sub-target multisets, so every enumeration count is zero). At $n = 9$, the 147 sub-target multisets partition as three with product exactly 7776 (table 10), 24 strictly in $(7776, 8748)$ (table 11), and the remaining 120 with product < 7776; the 2085 D_n-orbit reps partition consistently. For $n \geq 10$ the search is infeasible; see § 5.4.

n	M_n	multisets $< M_n$	D_n orbit reps	cand. good cycles	partial rule tables	rej. by pruning	rej. by verifier	C1+C2 rehash (16-char)
3	8	0	0	0	0	0	0	`61edaeb9e65bef43`
4	24	1	1	†	†	†	†	`4b667486854d5c46`
5	96	5	6	†	†	†	†	`60b039535065125b`
6	288	12	22	†	†	†	†	`d92808e8949d0dbc`
7	864	27	77	†	†	†	†	`6b84b65ca87a88fd`
8	2592	59	343	†	†	†	†	`91364b0e5c7bae90`
9	8748	147	2085	†	†	†	†	`1caca3ee3357c333`

Notes. † = populated from the pre-generated rejection-stream bundle shipped at `artifacts/rejections/` in the release tag (appendices C.10 and C.11); a referee can either replay that bundle or regenerate it by running the driver. The C1+C2 counts and rehashes in the shipped manifest already gate the deterministic portion. A driver cross-check hash (independent verifier re-implementation, appendix C.5) appears alongside in `coverage_manifest.json`. Wall-clock and hardware footprints per n are reported in appendix C.8. $n \geq 10$ omitted; see § 5.4.